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The human brain reactivates context-specific past information at event boundaries of naturalistic experiences

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Although we perceive the world in a continuous manner, our experience is partitioned into discrete events. However, to make sense of these events, they must be stitched together into an overarching narrative—a model of unfolding events. It has been proposed that such a stitching process happens in offline neural reactivations when rodents build models of spatial environments. Here we show that, while understanding a natural narrative, humans reactivate neural representations of past events. Similar to offline replay, these reactivations occur in the hippocampus and default mode network, where reactivations are selective to relevant past events. However, these reactivations occur, not during prolonged offline periods, but at the boundaries between ongoing narrative events. These results, replicated across two datasets, suggest reactivations as a candidate mechanism for binding temporally distant information into a coherent understanding of ongoing experience.

Reading a scientific paper is a complicated task. The paper you are about to read is composed of a stream of words that convey a detailed, interconnected narrative that you will try to make sense of. Although you may read through this narrative continuously, your brain will process the narrative by parcelling it into discrete chunks of information, termed 'events,'[1] with associated neural representations ('event models'[2,3]). The transitions between events, termed 'event boundaries,' have been suggested to occur when shifts in the content of incoming information will be detected, at which point the current event model will be encoded into memory and a new event model will be formed[1,2,4,5]. This could happen, for example, in the transition between paragraphs.

However, understanding a narrative requires more than just parcelling the ongoing stream of information into events and storing them into memory. Following an ongoing narrative requires relations to be drawn between each current event and contextually relevant past events (for example, linking results presented in the Results section with their related research hypothesis presented in the Introduction section). How, then, could these events be pieced together? It has long been proposed that remote past information could be integrated with incoming sensory information[6–8]. Accordingly, it has been suggested that event models of past events can assist in the interpretation of incoming information[5,9–12]. However, the mechanism underlying this process is currently unknown.

One possible mechanism that may underlie this integration is termed 'replay,' and was originally seen in rodents. Here, sequences of cell firing patterns that are observed during past events are rapidly recapitulated during pauses in behavior. Replay has been studied predominantly offline, during periods of rest or sleep[13,14], but it can be triggered by salient behavioral events, for example, when an animal completes a trial[15] or receives a reward[16]. These situations could be analogous to event boundaries, as they mark contextual shifts in the ongoing experience. This hypothesis is supported by human imaging studies, demonstrating that the representations of just-ended events are reactivated at event boundaries[17,18] (due to the lack of temporal resolution to infer a sequentially ordered reinstatement, the term reactivation, rather than replay, is commonly used).

Event segmentation and replay also engage similar brain regions. Increased activity at event boundaries during the processing of a

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narrative has been associated with regions including the hippocampus[19–21] and nodes of the default mode network (DMN[22], mainly the precuneus, angular gyrus/superior lateral occipital cortex (Ang/LOC) and superior/middle temporal gyrus[5,23–25]). Similarly, replay and sharp wave ripples are associated with increased activity in hippocampus and in the DMN in both humans and animal models[26–30].

It has been proposed that both replay in rodents and reactivation in humans encode past events into memory, as well as inform future decisions[13,31]. But, notably, offline replay in rodents is not limited to playing a veridical recording of experience. Instead, it can make inferences—piecing together several past events, even remote ones, as if building, and sampling from, a model of the world[13,32–34]. Similarly, in humans, replay can reorganize and reorder events to fit with a prelearned task model[35,36]. It is therefore possible that, as an experience is unfolding, reactivation could underlie the inference of relations between current and past events, and continuously bind related event models as the narrative is built progressively.

Replay studies have, so far, used controlled experiments with regularly repeated trials, which make it hard to assess the integration of events across time, since repeated events would have very similar representations. Yet, the integration of events across time can be measured in more naturalistic experiments that have a continuously evolving narrative: here, each event is unique and should have a more distinct representation. In such experiments, we would expect to find that reactivation at event boundaries would contain not only representations of the event that has just ended[17] but also representations of unique past events, particularly those most relevant to the understanding of the current narrative stage (for example, you will gain a better understanding of the current hypothesis if, at the relevant event boundary, your brain replays the hypothesis along with supporting information from previous paragraphs).

We therefore aimed to detect brain regions in which, not only the event that has just ended is reactivated at event boundaries, but also more temporally remote events that are relevant to the understanding of the present situation. To this end, we analyzed two naturalistic datasets, where participants either viewed a movie or listened to a story while undergoing a whole-brain functional magnetic resonance imaging (fMRI) scan. We examined these datasets using a new analytic method designed to detect reactivated representations of remote events at event boundaries. Using this method, we indeed revealed reactivation in the posterior hippocampus, the precuneus/retrosplenial cortex and the Ang/LOC, and replicated these results across datasets. We further demonstrated that this reactivation is apparent at both a local, voxel scale, and also at a mesoscale. Finally, we found that the reactivated representations in the precuneus/retrosplenial cortex were of semantically relevant past events, demonstrating a specific integration of information that is needed for the understanding of each current narrative stage. We therefore propose that reactivation is a constructive mechanism that builds our understanding of evolving experiences by selectively binding relevant pieces of information across time.

Results

Can reactivation at event boundaries integrate current, ongoing experiences with temporally remote past information? To answer this question, we used fMRI data acquired from participants engaged in a naturalistic narrative (movie or story). Reactivation entails the expression of past scene representations (voxel activity patterns) at event boundaries. Therefore, to detect reactivation in these fMRI data, we looked for correlations between fMRI representations at event boundaries (scene transitions) and representations of remote scenes (preceding the just-ended scene) using representation similarity analysis[37]. However, due to the low temporal resolution of fMRI data, representations at event boundaries are probably correlated with the representations of the immediately neighboring scenes. If these scenes are similar to other scenes, then correlations between event boundaries and remote scenes may in fact reflect between-scene correlations, rather than reactivation of remote information. This problem can be solved because between-scenes correlations are symmetric in time (past/future; Fig. 1b). We therefore defined a 'reactivation index' that cancels out symmetric similarities by contrasting past with future correlations. The results obtained from this reactivation analysis were replicated across two independent fMRI datasets (Sherlock dataset[38], 21st year dataset[39]; Table 1). We demonstrate these results using reactivation within the brains of single participants, and also using the similarity in reactivation between the brains of different participants (potentially measuring reactivation of representations at a mesoscopic scale). Finally, we used the overlap in words between scenes to test whether the reactivated scenes are those that are most relevant (semantically similar) for the understanding of the current scene.

See Fig. 1 for a schematic of the analysis pipeline and Methods for full methodological details, simulations and experimental constraints of our analytic methods.

Reactivation of remote past events at event boundaries

We first tested for the reactivation of remote events within a single brain. To this end, we looked for brain regions where representations at event boundaries are more similar to representations of past scenes compared with future scenes (significantly positive reactivation index). Visual inspection of the results, as presented in Fig. 2a, revealed that brain regions showing consistent effects across datasets included the bilateral precuneus/retrosplenial cortex, Ang/LOC and posterior hippocampus. These regions were consistently observed in each of the two datasets and in a mega-analysis that pooled the participants of the two datasets together (the top two rows of Fig. 4a display the overlap between these regions across datasets. The statistical quantification of this replication will be presented in Reactivation is specific to event boundaries and Fig. 4b,c). No regions showing the opposite effect, of a significantly negative reactivation index, were revealed. All maps have been cluster-corrected for multiple comparisons across the entire brain ($P < 0.005$, permutation test).

Note that, although regional variations between the group reactivation maps were also revealed, we refrain from interpreting results that do not show a qualitative replication across the two independent datasets. This is because such differences could be attributed to any of the design- or processing-related specifics of the individual datasets (Table 1; Methods).

Reactivated representations have structure at the mesoscale

Replay in rodents is typically studied at the single-cell resolution. The human reactivation results presented so far were voxel-specific, which is the highest possible spatial resolution when using fMRI. However, naturalistic fMRI findings indicate that the neural representations of scenes are shared across participants[40–42]. As there is no accurate correspondence between single voxels across brains (due to inter-brain anatomical and functional variability and image preprocessing constraints), it is likely that between-participant analyses do not sample the exact same voxels across brains, but rather voxels that are in some anatomical vicinity to each other. The findings of shared scene representations across brains therefore demonstrate that the representations of complex naturalistic events can be expressed at coarser scales, including mesoscopic scales, in which functional anatomy is consistent across participants. We were therefore interested to study whether reactivation is expressed at a mesoscopic resolution. Unlike in previous studies, we tested not whether mere scene representations are shared, but rather whether the reactivated representations observed at event boundaries within a single brain would also be shared across participants. To this end, scene representations extracted from one brain were correlated with the averaged event boundary representations across the remaining participants of the same dataset.

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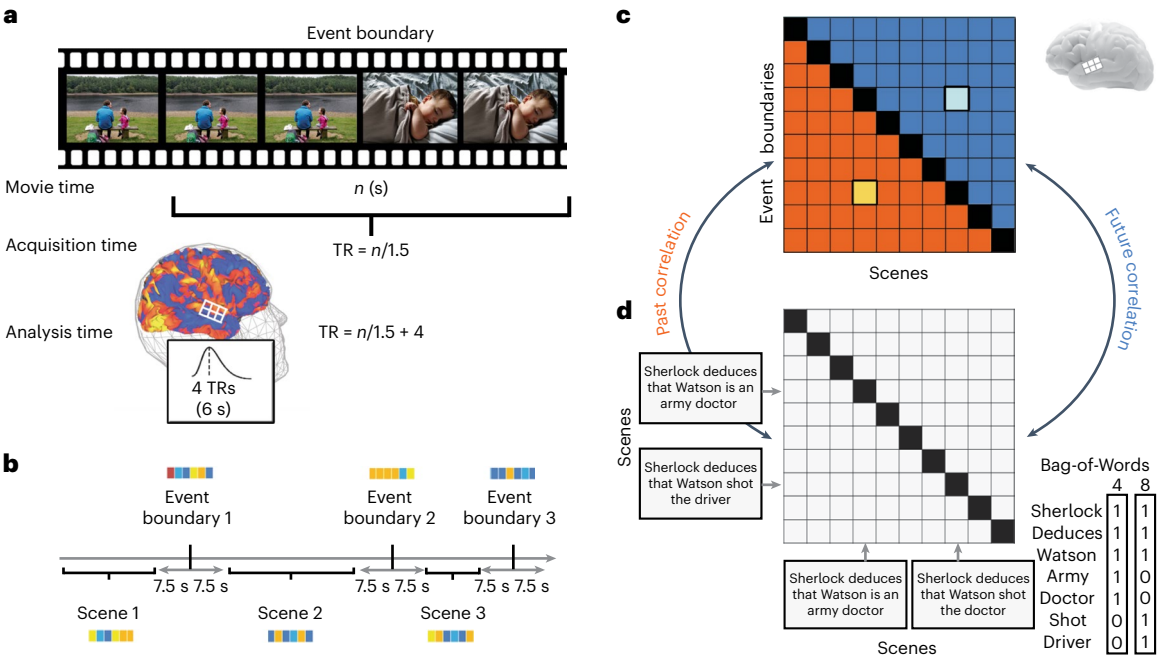


Fig. 1 | Analysis pipeline. a, Participants watched a movie or listened to a story while being scanned. We defined event boundaries as TRs in which scenes transitioned (TR = 1.5 s). We next shifted these TRs by 6 s (four TRs) to account for the hemodynamic delay. **b**, For each sphere in the brain, we extracted fMRI representations of event boundaries (black vertical lines). To separate the BOLD signals of event boundaries and scenes, we defined scene representations as the average representation across all within-scene TRs that are five TRs (7.5 s) remote from adjacent event boundaries. **c**, For each sphere, we created a similarity matrix by cross-correlating the representations of event boundaries and scenes. The lower/upper triangular parts of the matrix contained the similarities between event boundary representations and the representations of preceding/following scenes (past/future parts of the matrix, orange/blue color, respectively). We removed the main diagonal to measure only reactivation of temporally remote scenes. Note that if similarities between event boundaries and scenes merely reflected scene similarities, this similarity would be reflected in both the past and future parts of the matrix. For example, if the similarity between event boundary 8 and scene 4 (entry (8,4), bright orange) in fact reflected the similarity between scene 8 and scene 4, then the same similarity should be evident in the future part of the matrix (entry (4,8), bright blue). To control for this possibility, we defined a reactivation index as the difference between the means of the past and future parts of the matrix, thus cancelling out any symmetric scene similarity effects. **d**, Bag-of-Words analysis. We represented each scene according to the occurrences of words, from across the entire narrative, that were used in that scene (illustrated for scenes 4 and 8). We then cross-correlated these scene representations to create a context similarity matrix. Next, we correlated the past/future part of the neural similarity matrix with the past/future part of the context similarity matrix, respectively. Finally, we subtracted the future correlation coefficient from the past correlation coefficient, thus controlling for scene similarities. See Methods for the variations of this method used for different datasets and analysis method.

Table 1 | Details of datasets

	Sherlock	21st year
Number of participants	17	25
Stimulus type	Audiovisual movie	Auditory story
Genre	Mystery	Drama
Stimulus structure	A single main narrative	Two alternating narratives (scenes 1–30), which join into a single narrative at the final part of the story (scenes 31–45)
Total length	48 min	56 min
Number of scenes	48 (plus two audiovisual cartoons)	45
Median scene length (range)	51 (12–184.5) s	67.5 (60–84) s

Visual inspection of the results, as depicted in Fig. 3a, demonstrates that the between-participant analysis indeed revealed reactivation in the bilateral precuneus/retrosplenial cortex, Ang/LOC and bilateral posterior hippocampus, similar to the effects seen in the within-participant analysis (Fig. 2a; the bottom two rows of Fig. 4a display the overlap between these regions across datasets. The statistical quantification of this replication is presented in Reactivation is specific to event boundaries and Fig. 4b,c). For the 21st year dataset, these results were equally strong when only the congruent parts of the similarity matrix were analyzed, but not if a time (past/future) × congruency (same/different narrative) interaction was used as a reactivation index (Supplementary Fig. 1; Discussion). Similar results were also obtained for both datasets using the more typical scene-based between-participant analysis, in which the representations at event boundaries (rather than scene representations) extracted from one brain were correlated with the averaged scene (rather than event boundary) representations across the remaining participants of the same dataset (Supplementary Fig. 2, left column).

We interpret the greater correlations between representations at event boundaries and past scenes (compared with future scenes) as reactivation of past information. However, a possible confound to this interpretation is a case where event representations are rapidly 'flushed' after event boundaries[43]. If this is true, our effects may be due to representations at the event boundaries resembling the preflush representation more than the postflush representation. This can be addressed by excluding, not only the preboundary event, but also the postboundary event from the analysis (as already done in the within-participant analysis to control for autocorrelation-related biases). Doing so yielded results that are highly similar to the original ones (Supplementary Fig. 3), suggesting that flushing dynamics do not drive our effects.

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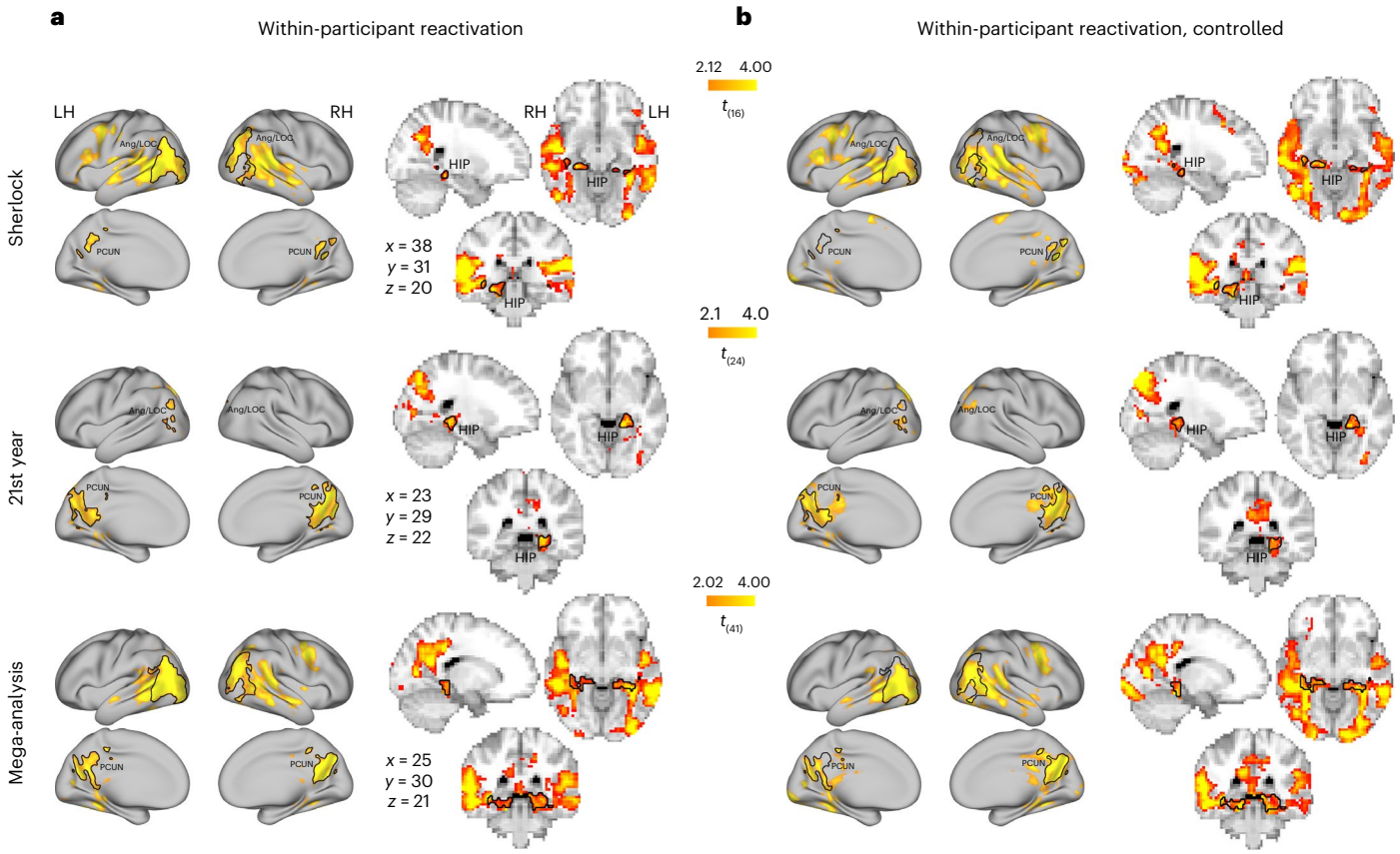


Fig. 2 | Within-participant reactivation of remote past events at event boundaries. a, Within-participant reactivation, where reactivation indices were derived from event boundary and scene representations that were extracted from the same brain. **b**, Controlled within-participant reactivation, where the within-participant reactivation indices at event boundaries were contrasted with those computed for control timepoints. Rows depict within-dataset group maps of the Sherlock dataset (first row), the 21st year dataset (second row) and a mega-analysis pooling together the two datasets (third row). Maps are presented on both flat cortical surface and on three-dimensional slices (dataset-specific MNI coordinates are depicted). Across all datasets, reactivation of temporally remote past events (as reflected in positive *t* values) was consistently found in the bilateral precuneus/retrosplenial cortex, the Ang/LOC and hippocampus. To illustrate the resemblance in results between the uncontrolled and controlled analyses, we superimposed ROIs based on the uncontrolled within-participants analyses of each dataset (black contours in **a**) on the corresponding controlled reactivation maps (**b**). LH, left hemisphere; RH, right hemisphere. All maps were created using two-sided *t*-tests and were cluster-corrected for multiple comparisons across the entire brain ($P < 0.005$).

Thus, our results so far reveal a set of brain regions in which remote scene representations are reactivated reliably at the event boundaries of naturalistic fMRI data in humans. The reactivated representations are evident not only within a single brain, but are also shared across brains, providing evidence that the reactivated representations have structure at the mesoscopic scale.

Reactivation is specific to event boundaries

We have so far shown reactivation of past events at event boundaries, but is this effect specific to event boundaries? To test this, we repeated the reactivation analysis, but this time contrasted the reactivation indices at event boundaries with those calculated for control timepoints that are adjacent to the event boundaries (Supplementary Fig. 4; Methods). As presented in Fig. 2b and Fig. 3b (see Supplementary Fig. 2 for results of the scene-based between-participant analysis), this analysis revealed effects at a very similar set of regions compared with the original, uncontrolled reactivation analysis (Figs. 2a and 3a and Supplementary Fig. 2). Indeed, overlaying the regions of interest (ROIs) defined from the uncontrolled analyses onto the maps resulting from the controlled analyses revealed a substantial overlap. This was true across datasets and analysis methods, attesting to a robust and specific effect of reactivation of remote past information to event boundaries.

The specificity of reactivation to event boundaries in the precuneus/retrosplenial cortex, posterior hippocampus and Ang/LOC is clearly evident by examining the reactivation timecourses in these regions (the reactivation index at each repetition time (TR) around the event boundary; Methods), presented in Fig. 4 (see Supplementary Fig. 5 for results of the scene-based between-participant analysis). Across these brain regions, the reactivation timecourses peaked 6–9 s after event boundaries (vertical red lines), in a timecourse compatible with a blood-oxygen-level-dependent (BOLD) response.

Of importance, quantifying the effects observed in these ROI-specific reactivation timecourses also allowed us to test statistically whether the same brain regions are indeed detected across the two independent replications of our results. We conducted a direct cross-validation across datasets (Sherlock and 21st year) by defining ROIs in one dataset and testing the magnitude of their reactivation signals in the other dataset using the same analysis method (Fig. 4). The tested brain regions showed highly replicable effects across these two independent datasets, using either analysis method, as reflected in the test-specific *P* values (Supplementary Table 1, the only exception is the hippocampus in the within-participant analysis of the Sherlock dataset, $P = 0.1$). Meta-analysis of *P* values across datasets confirmed these results (within-participant analysis: precuneus/retrosplenial cortex: $\chi^2_{(4)} = 27.63$, $P < 0.001$; posterior hippocampus: $\chi^2_{(4)} = 10.14$, $P = 0.038$; Ang/LOC: $\chi^2_{(4)} = 21.36$; $P < 0.001$, between participant analysis: precuneus/retrosplenial cortex: $\chi^2_{(4)} = 27.63$, $P < 0.001$; posterior hippocampus: $\chi^2_{(4)} = 23.47$, $P < 0.001$; Ang/LOC: $\chi^2_{(4)} = 27.63$; $P < 0.001$,

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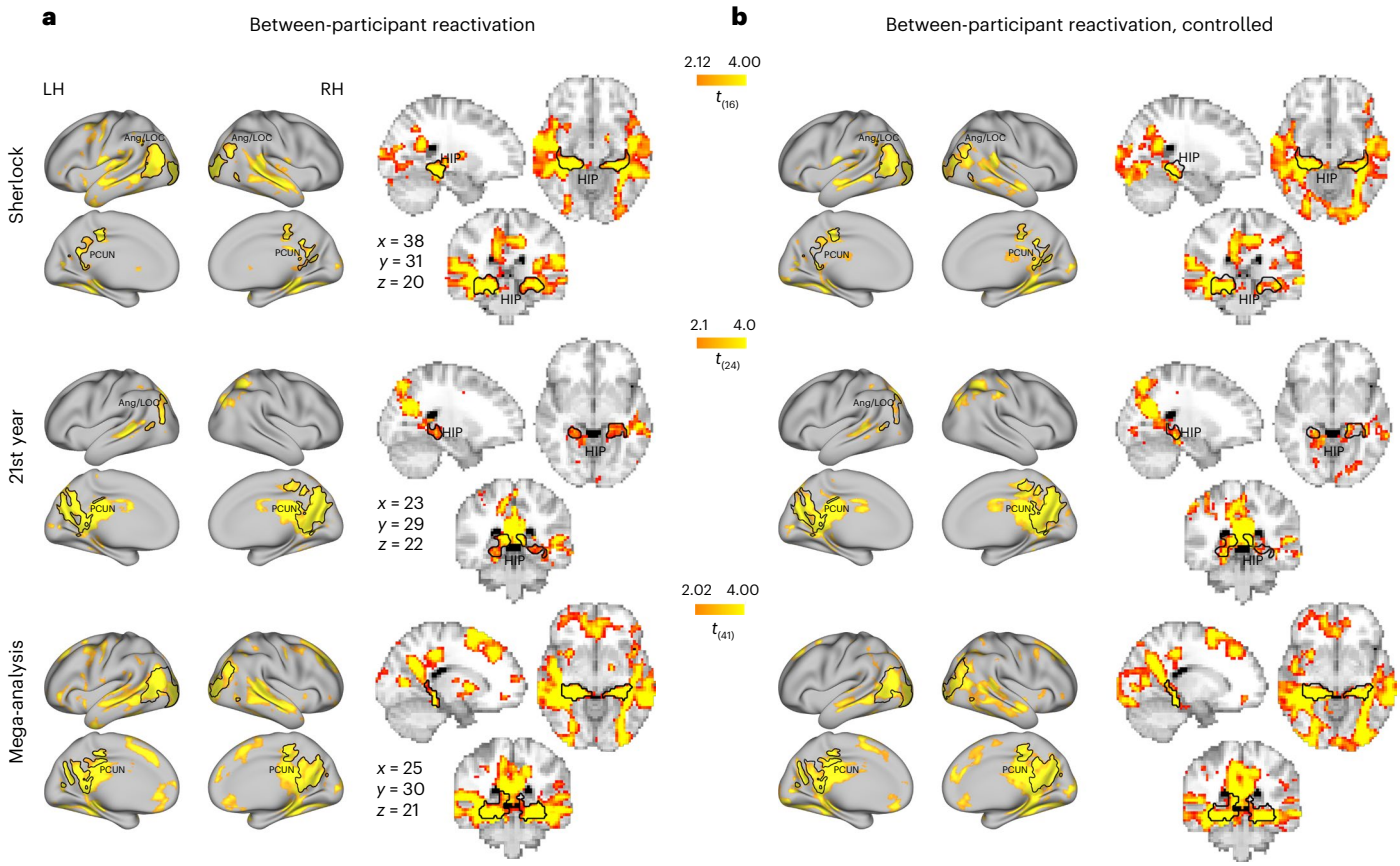


Fig. 3 | Between-participant reactivation of remote past events at event boundaries. a, Between-participant reactivation, where reactivation indices were derived from the scene representations of one participant and the event boundary representations from all other participants of the same dataset. **b**, Controlled between-participant reactivation, where the between-participant reactivation indices at event boundaries were contrasted with those computed for control timepoints. Rows depict within-dataset group maps of the Sherlock dataset (first row), the 21st year dataset (second row) and a mega-analysis pooling together the two datasets (third row). Maps are presented on both flat cortical surface and on three-dimensional slices (dataset-specific MNI coordinates are depicted). Across all datasets, reactivation of temporally remote past events (as reflected in positive *t* values) was found consistently in the bilateral precuneus/retrosplenial cortex, the Ang/LOC and hippocampus. To illustrate the resemblance in results between the uncontrolled and controlled analyses, we superimposed ROIs based on the uncontrolled between-participant analyses of each dataset (black contours in **a**) on the corresponding controlled reactivation maps (**b**). All maps were created using two-sided *t*-tests, and were cluster-corrected for multiple comparisons across the entire brain ($P < 0.005$).

Fisher's test, $\alpha = 0.017$ for all tests). Very similar results were obtained in a cross-validation analysis across datasets using the scene-based between-participant analysis (precuneus/retrosplenial cortex: $\chi^2_{(4)} = 27.63$, $P < 0.001$; posterior hippocampus: $\chi^2_{(4)} = 23.47$, $P < 0.001$; Ang/LOC: $\chi^2_{(4)} = 25.43$, $P < 0.001$, Fisher's test, $\alpha = 0.017$ for all tests; Supplementary Table 1 and Supplementary Fig. 5). The reactivation effect is therefore not only specific to event boundaries, but also highly replicable across datasets.

Reactivation is selective for relevant past events

Our results so far indicate a robust and reliable reactivation of remote past events, which is specific to event boundaries of ongoing naturalistic experience. But is there also specificity to the content of reactivated information? Is relevant past information reactivated more than irrelevant past information? To test this hypothesis, we constructed a 'Bag-of-Words' model of the text in each narrative (the scene descriptions and dialogues of the Sherlock movie and the full 21st year story). Each scene was represented by a vector, whose elements represented the number of occurrences of each word. For example, the Bag-of-Words vector for this paragraph would have a three in the element that represents the word 'vector' (see illustration at the bottom right of Fig. 1d). Under the assumption that similar contexts share similar words, by correlating these Bag-of-Words representations across scenes, we created a context similarity matrix for each experimental stimulus (Sherlock movie description and 21st year story; see illustration in Fig. 1d and similarity matrices in Supplementary Fig. 6b,d). Next, for each sphere in each participant's brain, we correlated the neural event boundary × scene similarity matrix with the dataset-specific Bag-of-Words scene-similarity matrix: for the Sherlock dataset, we correlated the past and future parts of the matrices separately and calculated a time (past–future) main effect. For the 21st year dataset, which contained two alternating narratives, we computed correlations between the past congruent (same narrative), past incongruent (different narrative), future congruent and future incongruent entries of the matrices, and computed a time × congruency interaction (Methods).

In the Sherlock dataset, a wide network of brain regions (including precuneus/retrosplenial cortex, Ang/LOC and hippocampus) showed a positive modulation of fMRI reactivation by context similarity across analysis methods (Fig. 5a). The 21st year dataset showed weaker effects (as expected due to methodological constraints; Methods), which replicated across datasets only in the precuneus/retrosplenial cortex (whole-brain corrected in the between-participant, but not the within-participant analysis; Fig. 5b). This effect in the precuneus/retrosplenial cortex was also evident in a mega-analysis across datasets (whole-brain corrected in the between-participant, but not the within-participant analysis; Fig. 5c. For additional results on the entire context similarity matrices see Supplementary Fig. 6; Methods). Significant results were also obtained using the scene-based

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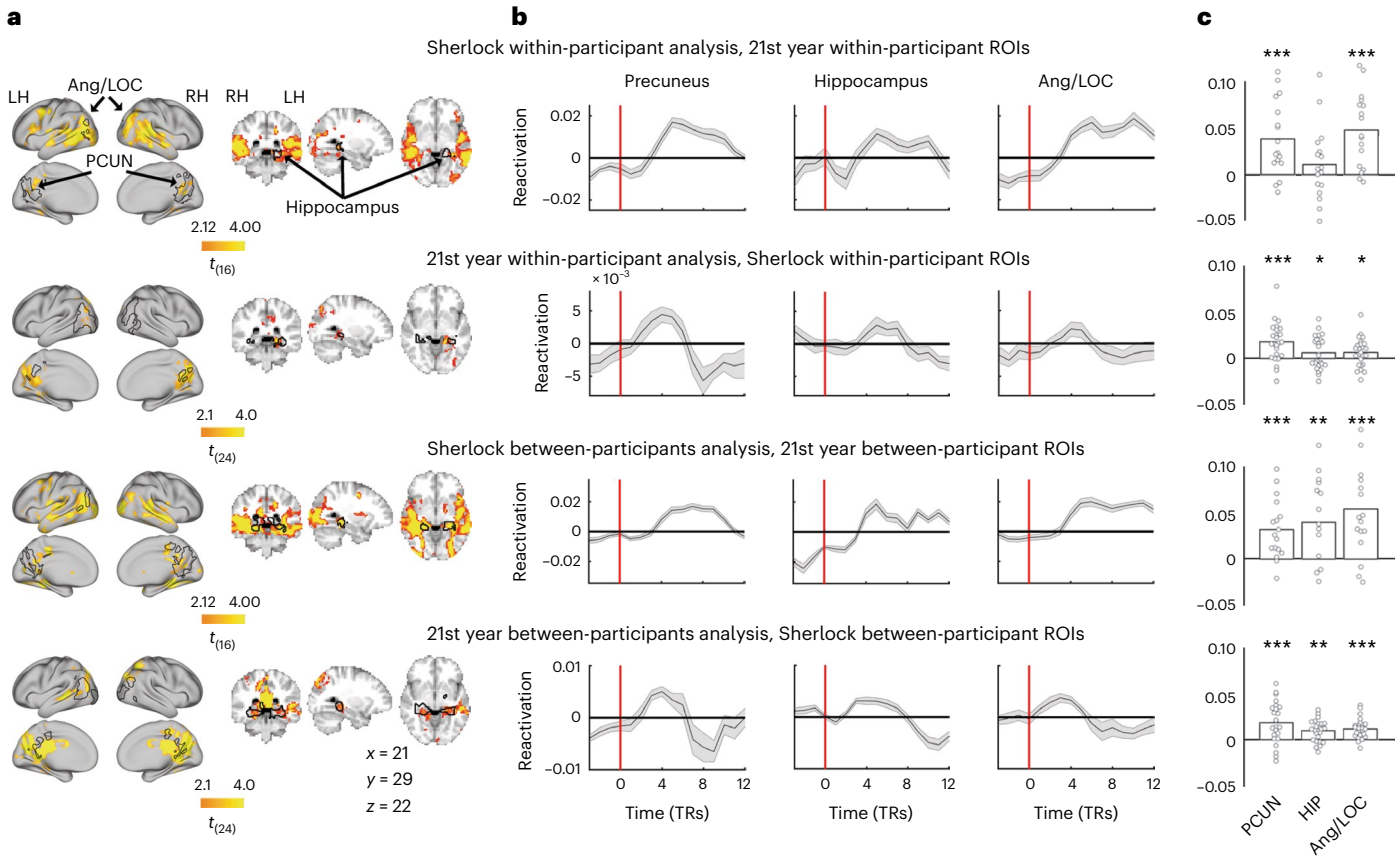


Fig. 4 | Reactivation is specific to event boundaries—replication across datasets. a, Group maps of each dataset (Sherlock and 21st year) superimposed with the ROIs defined from the other dataset, using the same analysis method (within- or between-participant analysis). For example, in the first row, the within-participant reactivation group map of the Sherlock dataset (Fig. 2a, 1st row) is superimposed with ROIs defined from the within-participant reactivation group map of the 21st year dataset (Fig. 2a, 2nd row). **b**, Group-averaged reactivation indices derived from the ROIs in **a** are presented for the timeperiod of 3 TRs before event boundaries until 12 TRs after event boundaries, uncorrected for the HRF delay. Event boundaries are depicted as red vertical lines. Gray error bands represent s.e.m. **c**, ROI analysis used to quantify the significance in **b**. Reactivation indices are derived from each of the datasets and ROIs presented in **a** and TRs 1–6 (minus baseline) presented in **b**. Single-participant values are represented as circles and group means are represented as bars; $n = 17$ and 25 participants for the Sherlock and 21st year datasets, respectively, see Supplementary Table 1 for exact P values. *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$, two-sided permutation tests.

between-participant analysis (Supplementary Fig. 7). These results suggest that, in the precuneus/retrosplenial cortex, reactivation at event boundaries is selective for remote past events that are relevant (have similar context) to the current situation.

While it is encouraging that most of these effects survive whole-brain correction, the critical question is whether the same voxels that reactivate past scenes on average (Figs. 2 and 3 and Supplementary Fig. 2), also do so selectively. To test this, we used an independent ROI analysis. We defined precuneus/retrosplenial cortex ROIs, separately from each dataset (Sherlock and 21st year) and analysis method (within and between participant) in the reactivation analysis (Fig. 2a and Fig. 3a), and tested the context specificity within each ROI in the same dataset and using the same analysis method (Fig. 5 and Supplementary Fig. 7). precuneus/retrosplenial cortex ROIs showed significant context-specific reactivations with each approach and dataset (Sherlock within-participant analysis: $t = 5.98$, $P < 0.001$, Sherlock boundary-based between participant analysis: $t = 4.86$, $P < 0.001$, Sherlock scene-based between participant analysis: $t = 10.55$, $P < 0.001$, 21st year within-participant analysis: $t = 1.92$, $P = 0.02$, 21st year boundary-based between participant analysis: $t = 1.51$, $P = 0.039$, 21st year scene-based between participant analysis: $t = 2.11$, $P = 0.019$, one-tailed permutation tests, see Supplementary Table 2 for means, s.e.m. and effect sizes). A meta-analysis conducted across the two independent datasets for each analysis method confirmed a robust replication of this effect (within-participant analysis: $\chi^2_{(4)} = 21.64$, $P < 0.001$; boundary-based between participant analysis: $\chi^2_{(4)} = 20.3$, $P = 0.001$; scene-based between participant analysis: $\chi^2_{(4)} = 21.36$, $P < 0.001$, Fisher's test). Mega analyses, pooling together all participants from the two datasets also confirmed these results (all $P < 0.001$; Fig. 5c and Supplementary Fig. 7c). Thus, the reactivation seen in the precuneus/retrosplenial cortex at event boundaries (Figs. 2 and 3 and Supplementary Fig. 2) is selective for relevant past information.

Discussion

How do we make sense of our ongoing experiences? Our findings propose that past information that is relevant for the comprehension of the current situation is reactivated at event boundaries. We found evidence for reactivation of past events in the same brain regions (precuneus/retrosplenial cortex, posterior hippocampus, Ang/LOC) across two independent datasets that are vastly different from one another (Table 1), suggesting that this reactivation represents a general attribute of the processing of ongoing experiences. These results, observed both within and between individuals, demonstrate that this reactivation exists both at the local (voxel-specific) scale and at the mesoscale (Figs. 2 and 3). We further established that the observed reactivation is specific not only to event boundaries (Fig. 4), but also to selected past events: the reactivation of events in the precuneus/retrosplenial cortex is modulated by semantic context, such that past events with a similar context

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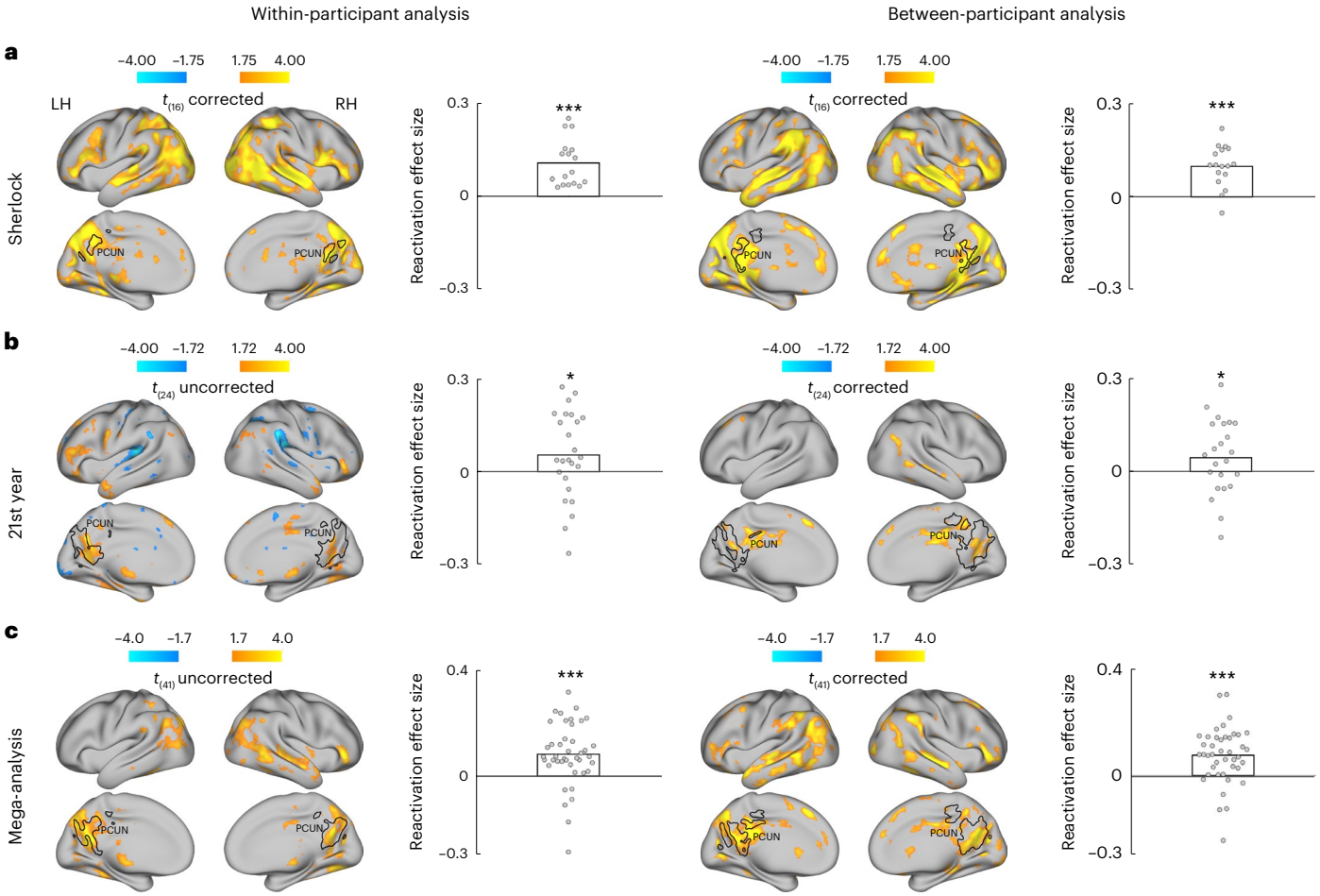


Fig. 5 | Relevant past events are preferentially reactivated at event boundaries. a–**c**, We measured the level to which reactivation is modulated by semantic context (as defined by a Bag-of-Words model; Fig. 1d) across the entire brain. Rows show results from the Sherlock dataset ($n = 17$ participants) (**a**), the 21st year dataset ($n = 25$ participants) (**b**) and a mega-analysis pooling together the two datasets ($n = 42$) (**c**). Columns depict results obtained using different analysis methods (within- and between-participant analysis). All maps were created using one-sided t-tests, and were cluster-corrected for multiple comparisons across the entire brain ($P < 0.005$), except for the within-participant analysis of the 21st year dataset (**b**, first column), and the within-participant mega-analysis (**c**, first column). Maps are projected onto a flat cortical surface. Across datasets and analysis methods, the precuneus/retrosplenial cortex showed a consistent positive modulation of reactivation by semantic context (scenes with semantic context similar to that of each event boundary were reactivated more than scenes with different semantic contexts). To confirm that the same regions that showed a reactivation effect also showed a significant modulation of reactivation by semantic context, we tested independent precuneus/retrosplenial cortex ROIs (black contours), defined separately for each dataset (Sherlock and 21st year) and analysis method (within- and between participant) in the reactivation analysis (Fig. 2a and Fig. 3a). We calculated the participant-specific reactivation effect sizes in these ROIs as the time (past–future) main effect/time × congruency (same/different story) interaction in the Sherlock and 21st year datasets, respectively (Methods). Single-participant effect sizes are represented as circles and group means are represented as bars (Sherlock within- and between-participant analysis, $P < 0.001$; 21st year within- and between-participant analysis, $P = 0.02$ and 0.039, respectively; mega analyses, $P < 0.001$, one-sided permutation tests). *$P < 0.05$; ***$P < 0.001$.

as the current event were more likely to be reactivated (Fig. 5). Taken together, these findings demonstrate that reactivation of past events at event boundaries is a selective mechanism that can piece together relevant parts of an ongoing experience.

Previous studies of replay have used experimental designs based on several repetitions of the same task[13,14,44]. Such experiments demonstrated that past knowledge (for example the prelearned structure of a maze) and current learning (for example, the position of a new reward location in the maze) can be combined through replay[33,44]. However, the documented replay was not selective for specific past events, but rather represented a whole knowledge structure, acquired across repeated trials. In other words, replay has been shown to reflect the assimilation of information into a previously acquired schema. Contrary to such overtrained experiments, in the naturalistic processing of a narrative, each event is experienced only once. Accordingly, we designed an analysis method that allowed us to treat events as unique pieces of information whose interrelations can be inferred as a narrative unfolds. Indeed, our observation of a selective reactivation of unique events that are semantically related to one another suggests the formation of relations between these individual events. Inferring these relations lies at the heart of understanding unfolding events, and we therefore suggest reactivation as the mechanism that constructs this relational understanding while events unfold.

Reactivation was observed in the hippocampus and in regions of the DMN. Indeed, hippocampal replay and sharp wave ripples have been documented simultaneously with activity in DMN regions in the human[27,28], primate[29] and rodent brain[26,30]. Replay and reactivation are commonly interpreted as supporting memory of past events and planning of future behavior[13,31]. The DMN is also commonly believed to be an 'offline' (resting-state) network, involved in remembering the past and envisioning the future, but to be disengaged from processing immediate external information[45,46]. However, in naturalistic experiments, the

DMN has been hypothesized to have an online function in interpreting ongoing information in light of previous knowledge[9,10,40,47], although the mechanism underlying the continuous binding of information has not yet been characterized. The naturalistic paradigm used here in combination with the mechanistic insights offered by our analytic approach indicate that, during an unfolding experience, the DMN processes online information by employing a reactivation mechanism that binds the present with related past events. The online nature of this computation can therefore construct our understanding of current events. Note that, it is not under dispute that reactivation or the DMN support offline functions such as memory or planning. The online-constructed links between events could build our understanding of the present, and simultaneously be consolidated into memory and inform subsequent decisions. We therefore suggest that reactivation in the DMN may do more than consolidating past events into memory or informing future decisions; it may serve as an information-binding mechanism that underlies the understanding of the present.

Not all regions of the DMN have been detected to reactivate remote past information in our results. However, both implicated nodes, the precuneus/retrosplenial cortex and the angular gyrus, have been highlighted as particularly likely to represent event models[4,5,48]. In addition, these two regions have been shown to have long timescales of neural dynamics[9,48,49], and were therefore suggested to be able to intrinsically retain past information throughout an uninterrupted movie viewing[9]. It is therefore possible that, at event boundaries, the precuneus/retrosplenial cortex and the angular gyrus can reactivate specific remote event models out of all past models they hold for the duration of the movie. The detection of neural signatures of event boundaries in these particular regions has been demonstrated to predict hippocampal response during movie watching[48]. It is therefore possible that, after the selection of events to be reactivated in these DMN regions, the hippocampus can bind these events with the current event model, resulting in an interconnected representation of an unfolding experience[7,50–53].

These regions were detected reliably in both datasets using a time main effect as a reactivation index, but not when using a time × congruency interaction in the 21st year dataset. This could be due to the fact that the 21st year story has an alternating narrative structure. At the end of each narrative-specific scene, the listeners probably predict that the following scene will be of the other (incongruent) narrative. It is therefore plausible that event boundaries will not only trigger reactivations that encode the content of the just-ended scene, embedded in its narrative. Event boundaries may also trigger reactivations of the incongruent narrative, to set the context for the predicted upcoming scene. If this is true, event boundaries would display reactivations of both narratives (congruent and incongruent with the just-ended scene), that will be evident in the time main effect but not in the time × congruency interaction. Future studies will be needed to investigate interaction effects in naturalistic stimuli that contain two narratives with an unpredictable structure.

We note that, of the three brain regions detected to reactivate past information, only the precuneus/retrosplenial cortex showed a context-selective modulation of this reactivation, which was reproducible across datasets. The retrosplenial cortex is indeed believed to process relational and contextual information needed for the consolidation of events into memory[54]. In addition, naturalistic experiments have repeatedly identified the precuneus as the main brain region involved in context processing[9,40,47] and in the representation of semantically central events (showing dense semantic relations to other events) during recall of a narrative[55]. It therefore seems reasonable that, as our results indicate, the precuneus/retrosplenial cortex has a unique contribution to the selection of contextually relevant events for reactivation during narrative processing. Nevertheless, our results also hint that other brain regions may take part in this process, as evidenced by the results obtained from the Sherlock dataset (Fig. 5a). Since these other brain regions were not replicated in the 21st year dataset, future work, using a larger number of datasets, will be needed to target the contextual modulation of reactivation in additional brain region.

It is also noteworthy that our analyses detected regions that reactivated past events, but no regions where representations at event boundaries were more similar to (and hence predictive of) future events. In theory, such predictive coding could have been detected in our analyses, since prediction and planning for the near future have been postulated to be integral to perception, and to be represented in event models[1,2,5]. Indeed, in the rodent literature, replay has been documented to inform immediate goal-directed choices[33,56,57] (but also see Gillespie et al.[58]). Future studies will be needed to characterize the conditions under which predictive coding could also be detected in narrative processing experiments in humans.

In conclusion, by the end of this paper, the reader's brain should have constructed an interconnected set of representations, relating our experimental hypotheses (presented in Introduction), our findings (presented in the Results), and our interpretations of these findings, as just described (we assume not all readers will have read the Methods section). We propose that hippocampus and DMN regions in the reader's brain formed this complex structure of knowledge via a selective reactivation of relevant information at event boundaries, such as endings of paragraphs. If this knowledge structure has indeed been formed during the reading, the reader should have been able make sense of this paper, and perhaps even remember it in the long run or make use of it in her/his own future work.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41593-023-01331-6.

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Methods

Overview

Here, we aimed to study reactivation of temporally remote events (that is, further than the just-ended event) at event boundaries of naturalistic stimuli. For this purpose, we analyzed two naturalistic datasets, where participants either watched a movie or listened to a story while undergoing a whole-brain fMRI scan. We measured reactivation as the correlation between fMRI representations at event boundaries (scene transitions; Fig. 1a), and representations of previous scenes (Fig. 1b). The lower/upper triangular parts of the resulting similarity matrix reflected similarities between each event boundary and earlier/later scenes (orange/blue, respectively; Fig. 1c).

We wanted to look for reactivation of remote scenes, so we removed the immediately preceding scene of each event boundary. However, one possibility is that each event boundary correlates with its immediately neighboring scenes, and these scenes correlate with other scenes. Thus, correlations between event boundaries and remote scenes may in fact reflect between-scene correlations, rather than reactivation of remote information. Yet, correlating between scenes results in a symmetrical similarity matrix, in which the same inter-scene correlations exist for both past scenes (scene eight versus scene four) or future scenes (scene four versus scene eight; Fig. 1c). It is therefore possible to control perfectly for this potential confound by computing a reactivation measure that contrasts the past with the future (thus cancelling out symmetric similarities). By computing this reactivation index using a whole-brain searchlight approach (namely, using a sphere centered on each fMRI voxel in each participant's data), we were able to test in which brain regions representations at event boundaries correlate with (that is, reactivate) past scene representations.

In this within-participant analysis, we calculated correlations between representations at event boundaries and scenes, extracted from a single brain. We also performed a between-participant analysis, in which representations of scenes were extracted from the brain of one participant and correlated with the averaged event boundaries representations extracted from the brains of the remaining participants of the same dataset. Note that, for the within-participant reactivation analysis, we needed to carefully control for autocorrelations in the raw data, since representations of event boundaries and scenes were extracted from the same fMRI signals. This problem will be discussed in Autocorrelations in within-participant analyses and does not apply to the between-participant analysis, which correlates representations between signals of independent participants.

Finally, to test whether reactivation is selective for scenes with relevant semantic context, we used the overlap in words between scenes to further characterize the observed reactivation (Bag-of-Words model; Fig. 1d).

Data

This study reports findings from two naturalistic fMRI datasets (Table 1): the Sherlock dataset[41] downloaded from Princeton University's DataSpace repository[38] and the 21st year dataset[40] downloaded from the Narratives repository[39].

Participants. Sherlock: 22 participants (10 female, age range 18–26 years) were recruited for the original study, of which 17 met the original study's inclusion criteria, and were therefore also analyzed in the current study.

21st year: 25 participants (14 female, age range 18–40 years) were analyzed in the original and current study.

All participants were right-handed native English speakers, and none had been exposed to the experimental stimulus before the scanning session.

Experimental stimuli. Sherlock: here we analyzed data from participants who watched the first 48 min of Episode 1 of the British Broadcasting Corporation (BBC)'s television series 'Sherlock'. The plot had a continuous structure, with varying scene-lengths (median 51 s, range 12–184.5 s). Scene timings were taken from the original study, where they were defined manually based on important shifts in the narrative[41]. The experimental stimulus was divided into two segments of 23 and 25 min, and a 30-s audiovisual cartoon (unrelated to the main stimulus) was presented before each segment.

21st year: participants listened to a 56-min-long story, which was segmented into scenes by design. These scenes had relatively constant lengths (median 67.5 s, range 60–84 s), and were separated by 4.5–6 s of silence. The narrative was composed of two unrelated storylines that alternated during the first 30 story scenes, and then merged into a unified storyline during the last 15 scenes.

fMRI acquisition. Sherlock: imaging was performed on a 3 T full-body scanner (Siemens Skyra) with a 20-channel headcoil. Functional images were acquired using a T2*-weighted echo planar imaging (EPI) pulse sequence (TR 1,500 ms; TE 28 ms; flip angle 64°), each volume comprising 27 slices of 4 mm thickness, in-plane resolution 3×3 mm^2, FOV 192×192 mm^2), with ascending interleaved acquisition.

21st year: imaging was performed on a 3 T full-body MRI scanner (Skyra, Siemens) with a 20-channel headcoil. Functional images were acquired using the same parameters as in the Sherlock dataset. Anatomical images were acquired using a T1-weighted magnetization-prepared rapid-acquisition gradient echo (MPRAGE) pulse sequence (TR 2,300 ms; TE 3.08 ms; flip angle 9°; $0.86 \times 0.86 \times 0.9$ mm^3 resolution; FOV, 220×220 mm^2).

Data preprocessing. Sherlock: fully preprocessed images from the Sherlock dataset were downloaded and used. Preprocessing was performed in FSL (http://fsl.fmrib.ox.ac.uk/fsl), and included slice time correction, motion correction, linear detrending, temporal high-pass filtering (140 s cutoff) and coregistration and affine transformation of the functional images to a Montreal Neurological Institute (MNI) template brain. Functional images were resampled to 3 mm isotropic voxels.

21st year: raw images from the narrative dataset were downloaded and then preprocessed using FSL and inhouse Matlab code (v.2018a, Mathworks). Functional data were analyzed using FMRIB's expert analysis tool (FEAT, v.6). The following preprocessing steps were applied to each participant's data: motion correction using FMRIB's linear image registration tool (MCFLIRT) and brain extraction using BET and high-pass temporal filtering (120 s cutoff). Functional images were aligned to structural images initially using FMRIB's linear image registration tool (FLIRT)[59,60] and then optimized using Boundary-Based Registration (BBR)[61]. Structural images were transformed into MNI space using FMRIB's nonlinear image registration tool (FNIRT) and the resulting warp fields were applied to the functional images. Tissue-type segmentation was carried out using FAST to create white matter/cerebrospinal fluid (CSF) nuisance masks. To avoid the inclusion of gray matter voxels in these nuisance masks, these masks included only voxels identified as white matter/CSF with probability of 1, restricted by an anatomically based bounding box ($-44 < x < 42$, $-84 < y < 42$, $-4 < z < 34$ for white matter; $-42 < x < 38$, $-64 < y < 38$, $-22 < z < 28$ for CSF), and eroded by a sphere of radius 5 and 2 mm for white matter and CSF, respectively[62]. White matter and CSF timecourses were extracted for each functional scan, and their contribution to the BOLD signal, as well as the contribution of motion parameters, was later removed. Functional images were resampled to 3 mm isotropic voxels.

Across all analyses, unless otherwise specified, fMRI BOLD responses were shifted by four TRs (6 s) in relation to the experimental stimuli to account for the canonical hemodynamic response function (HRF) delay (Fig. 1a).

Data analysis

fMRI representations. To test reactivation across the entire brain, a searchlight approach was employed, with spheres of three voxel (9 mm)

radius confined by a standard MNI mask. Extracting representations (time-specific voxel patterns[63]) from each sphere necessitated the separation of fMRI BOLD signals of event boundaries from the signals of whole scenes. Therefore, as illustrated in Fig. 1b, in each participant, representations of event boundaries were defined as the voxel pattern in the single TR that ends each scene. Representations of scenes were defined as the average representation across all within-scene TRs that are five TRs (7.5 s) remote from adjacent event boundaries. The selection of a five-TR gap was based on the estimation that the canonical HRF peaks at 4–6 s (three to four TRs) of stimulus onset, hence a minimal gap was chosen as to avoid the HRF peak but also to allow sufficient scene TRs to be analyzed.

Note that within-scene averaging of representations is a common practice[40–42,64,65], which is consistent with the pattern of whole-scene reactivation at event boundaries seen in electroencephalography data[17]. This averaged scene representation ignores rapidly changing scene information and probably reflects more stable elements of the scene, or its gist, as believed to be captured by event models[4].

Due to the need to separate between the BOLD signals of event boundaries and scenes, five scenes from the Sherlock dataset that were shorter than ten TRs were excluded from further analyses, along with their corresponding event boundaries. In the same dataset, the two scenes corresponding to the unrelated cartoon added to the experimental stimulus were also removed, along with the final scene of each scan, which did not have enough TRs following it to account for the HRF delay.

Within- and between-participant similarity matrices. To detect similarities in representations that would indicate information reactivation, the Pearson correlations between event boundary representations and scenes representations were calculated (Fig. 1c). This yielded an event boundary × scene similarity matrix, in which the lower triangular part reflected correlations between each event boundary and its previous scenes, and the upper triangular part reflected correlations between each event boundary and its following scenes. The main diagonal of this matrix was removed, as it reflected similarities between each event boundary and its immediately preceding scene, rather than reflecting similarities with remote scenes.

Importantly, this similarity matrix was computed using two related analytic approaches: a within-participant analysis and a between-participant analysis. In the within-participant analysis, for each sphere in each participant, representations of event boundaries and of whole scenes were derived from a single brain, and correlated to create a within-participant (event boundary × scene) similarity matrix. Thus, this approach allowed the detection of reactivation of remote scene representations at event boundaries within a single brain.

However, under the assumption that brain activity is modulated by naturalistic stimuli in a similar fashion across participants, correlations between brains could also be calculated (intersubject correlations[66]). This approach has been implemented in the analysis of naturalistic data, demonstrating that representations of scenes have structure at a mesoscale in which functional anatomy is shared across participants[40–42]. Hence, reactivation too may have a structure at a mesoscale, and could be detected in the similarity between the representations of event boundaries in one brain and the (shared) representations of scenes averaged across all other brains.

However, it is also possible that the representations at event boundaries may be shared across brains. This previously unexplored possibility would suggest that the reactivated pattern itself, rather than the representations of scenes, is common across all participants. To test this hypothesis, for each sphere in each participant, representations of scenes were extracted and correlated with the same-sphere event boundary representations, which were averaged across all other participants of the same dataset. This resulted in an event boundary (group-averaged) × scene (of one participant) similarity matrix.

Note that, the brain regions showing reactivation would not necessarily be similar across these two types of analysis. First, the within-participant analysis should be equally sensitive for the detection of reactivation across all brain regions. This is unlike the between-participant reactivation, which could be detected only in brain regions in which representations are shared across participants. Second, compared with the within-participant analysis, the higher signal-to-noise ratio of the between-participants analysis, resulting from the averaging of signals across many brains, may allow an easier detection of areas that show a reactivation effect (thus resulting in stronger effects).

Furthermore, autocorrelations needed to be controlled for in the within-participant reactivation analysis, since representations of event boundaries and scenes were extracted from the same fMRI signals (as will be discussed in Autocorrelations in within-participant analyses). However, in the between-participant analysis, the representations of event boundaries were temporally independent of the scene representations (as they came from different brains), and therefore no autocorrelation issues existed.

All following analyses reported here were conducted using both the within-participant approach and the boundary-based between-participant approach. In addition, results from the scene-based between-participant analysis, using an event boundary (of one participant) × scene (averaged across all other participants) similarity matrix, are presented in the supplementary materials.

Finally, it is possible that greater correlations between representations at event boundaries and past scenes (compared with future scenes) representations reflect event representations that are rapidly 'flushed' after event boundaries[43]. This would entail that representations at the event boundaries resemble the preflush representation more than the postflush representation. To rule out this possibility, the between-participant analyses were repeated while excluding the postboundary events (in addition to the preboundary events). Note that this control analysis was not needed for the within-participant analysis, as the related events were already removed to control for autocorrelation-related biases (Autocorrelations in within-participant analyses).

Reactivation analysis. Correlations between event boundary- and scene-representations were meant to detect reactivation of remote scenes at event boundaries. However, it is also possible that these correlations would simply reflect similarities between scene representations, for two reasons: (1) due to the low temporal resolution of fMRI data, event boundaries are probably correlated with their surrounding scenes; (2) if each event boundary representation contains reactivation of its immediately preceding scene[17], the representation at the boundary would be similar to the scene representation. Since same-narrative scenes are likely to have shared features and, therefore, similar representations, the representation at the event boundary of a certain scene may be similar to remote scenes, due not to reactivation, but merely to between-scene similarities (the event boundary representation is similar to its neighboring scene representations, the neighboring scene representations are similar to other scene representations). However, between-scenes similarities necessarily yield a symmetric similarity matrix, in which the same inter-scene correlations exist for both past scenes (for example, scene eight versus scene four) or future scenes (for example, scene four versus scene eight; Fig. 1c). Therefore, to rule out the possibility that between-scene similarities were measured, rather than reactivation of remote scenes at event boundaries, a 'reactivation index' was used. This index was defined as the difference between the mean of the lower triangular part of the similarity matrix (past part), and the mean of the upper triangular part of the similarity matrix (future part):

$$\frac{n(n-1)}{2}\left(\sum_{i=2}^{n}\sum_{j=1}^{i-1}M_{ij}-\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}M_{ij}\right)$$

Where M is an $n \times n$ matrix, i and j are row and column indices, respectively.

This index controls for any effects of scene similarities, since it cancels out symmetric entries in the lower and upper triangular parts of the matrix, and thus isolates the asymmetric components of the matrix as a probe for reactivation.

For the 21st year datasets, we also repeated the between-participant reactivation analyses using only the congruent entries of the similarity matrix, and also using a reactivation index based on a time (past/future) × congruency (same/different narrative) interaction.

Thus, our analytic approach consisted of the following steps: for each sphere in each participant, event boundary and scene representations were extracted. An event boundary × scene similarity matrix was calculated, from which the main diagonal was removed. Correlation coefficients in the matrix were Fisher-transformed, and a reactivation index was then calculated and assigned to the anatomical coordinates of the sphere's center. This procedure was repeated for all brain spheres, creating a single-subject map. Within each of the Sherlock and 21st year datasets, single-subject maps were submitted to a voxel-wise two-tailed group t-test, creating a dataset-specific group map, thresholded at $P < 0.05$. These group maps were corrected for multiple comparisons using a permutation test, with the null distribution of the maximal cluster mass, as implemented in FSL's randomize, and thresholded at $P < 0.005$. Finally, the corrected maps were projected onto a template MNI brain using the connectome workbench.

ROI definition. Regions that showed consistent effects across datasets were detected visually. The four reactivation group maps (resulting from the Sherlock and 21st year datasets in the within- and between-participant analyses) were masked by relevant regions from anatomical atlases to define four sets of ROIs, as detailed below.

All ROIs were defined based on group maps in a statistically significant threshold, and after correction for multiple comparisons. Anatomical definitions of the precuneus and lateral occipital cortex (joining the superior and inferior subdivisions) from Harvard-Oxford cortical structural atlas were used to mask the group maps, creating dataset- and analysis-specific precuneus/retrosplenial cortex and Ang/LOC ROIs. All hippocampal subregions of the Julich histological atlas were merged to create an anatomical definition of the hippocampus, which was used to mask all group maps and create dataset- and analysis-specific hippocampal ROIs.

Time-specificity of reactivation

To test whether the observed reactivation is specific to event boundaries, representations at event boundaries must be contrasted with those of a control timepoint. But which control timepoint should be chosen? Event boundaries are, by definition, positioned between two adjacent scenes. A control timepoint that precedes an event boundary will be much closer to the past period (as defined by the boundary) and a control timepoint that follows an event boundary will be much closer to the future period. Due to autocorrelations in fMRI signals, the choice of each of these timepoints to serve as a control will bias our analyses (as detailed in Autocorrelations in within-participant analyses). This problem can be tackled by choosing both a past and a future control timepoint that are both equally distant in time from the event boundary, and thus have symmetric time-biases that will cancel out.

Therefore, a balanced control was defined as the averaged reactivation indices computed for two symmetrical timepoints on both sides of each event boundary. Contrasting the reactivation indices of event boundaries with those of the balanced controls was defined as:

$$\mathrm{Index}_{\mathrm{EventBoundary}} - \frac{1}{2}\left(\mathrm{Index}_{\mathrm{PastControl}} + \mathrm{Index}_{\mathrm{FutureControl}}\right)$$

Control timepoints should be chosen such that they would not include any contribution from their adjacent event boundaries, otherwise the experimental effects seen at the event boundaries would be diminished by the contrast. Given the sluggish nature of the BOLD response, control points that are ten TRs (15 s) distant from the event boundaries were chosen for the Sherlock dataset. This optimal choice, however, could not be made for the 21st year dataset, given its experimental design. Since most event boundaries are preceded by a scene of the same narrative and followed by a scene from a different narrative, the past and future controls would potentially be contaminated by congruency effects (higher reactivation indices for past versus future control timepoints). For this reason, control timepoints that are only five TRs (7.5 s) distant from the event boundaries were chosen for the 21st year dataset. Since scenes in this dataset were separated by three to four TRs of silence, these control timepoints were the furthest possible from the event boundary while not introducing any incongruent effects into the future control timepoint. Both control timepoints were therefore well within a 'congruent' BOLD response, thus eliminating unbalanced congruency effects. Yet this congruent response was shared with the event boundary itself. For this reason, the contrast between event boundaries and control reactivation indices, calculated for the 21st year dataset, was suboptimal.

The contrast between the reactivation index of event boundaries and control timepoints was calculated for each sphere in each participant's brain. Single-subject and group maps were created as described previously. For consistency, a controlled reactivation analysis was additionally performed with control timepoints that are five TRs away from event boundaries for the Sherlock dataset.

Reactivation across time. To visualize and measure the reproducibility of the observed reactivation effects across datasets (Sherlock and 21st year) and using the two analysis methods (within- and between-participants), the timecourses of reactivation were drawn. These timecourses were extracted from the previously described ROIs of brain areas that showed consistent reactivation effects across datasets: the bilateral precuneus/retrosplenial cortex, bilateral posterior hippocampus and the bilateral Ang/LOC.

Reactivation signals from these ROIs were extracted from independent datasets and using the same analysis method: ROIs defined from the Sherlock dataset were tested on the 21st year dataset, and vice versa. ROIs defined based on the within- and between-participant analysis were tested using the same analysis method. This procedure allowed the conduction of two cross-validation tests across the independent datasets using each of the two analysis method.

Statistical analysis. For each of these ROIs, and in each participant's data, reactivation indices were extracted for single TRs around the event boundaries, in the manner described in Reactivation analysis. However, the current reactivation indices were uncorrected for the HRF delay, to demonstrate the true temporal correspondence between the reactivation indices and event boundaries. The time window chosen for this analysis included 3 TRs before event boundaries and 12 TRs after the event boundaries. Since fMRI autocorrelations are substantial in short timeperiods (Autocorrelations in within-participant analyses), this analysis was restricted to event boundaries that originated from scenes with a minimum of 20 TRs. To provide a visual demonstration of the changes in reactivation across time, the reactivation indices were averaged across events within participant, and then averaged across participants of the same dataset.

To test the magnitude of reactivation at event boundaries across datasets and analysis methods, the following procedure was employed: the TR-specific reactivation values were divided into a baseline (the HRF-uncorrected event boundary and the TR that precedes it) and signal (TRs 1–6 after the HRF-uncorrected event boundary, to account for the HRF delay). For each participant, the averaged reactivation value of the baseline was subtracted from the averaged reactivation value of the signal. This resulted in a single reactivation index for each participant and dataset/analysis method. Note that this index is different

from the original reactivation index, which measured the magnitude of reactivation at a specific TR (the HRF-corrected event boundary). Cohen's D effect sizes were calculated by dividing each group's mean by the groups' s.d.

Group statistics were carried out using permutation tests. Single-subject indices were submitted to a one-sample *t*-test, and the group level *t*-value was defined as the statistic for the permutation test. Under the null hypothesis of no reactivation (a flat signal), the sign of single-subject reactivation indices could be flipped. Thus, for each dataset/analysis method, reactivation indices of single participants were flipped randomly, and a random group *t*-value was calculated. This procedure was repeated 1,000 times, resulting in 1,000 random *t* values that constructed the null distribution. The position of the true test statistic in relation to the null distribution was used to determine the two-tailed *P* value of the test.

To test the reproducibility of results across datasets, for each ROI separately, the two dataset-specific *P* values resulting from the above described permutation tests were meta-analysed using Fisher's method[67,68]. To correct for multiple hypotheses testing across the three ROIs (precuneus/retrosplenial cortex, hippocampus and Ang/LOC), the alpha level was adjusted to 0.017 based on the conservative Bonferroni correction.

Context specificity of reactivation. Whole-brain analysis. Is reactivation of remote past information context-selective? In other words, would past scenes that share a similar context with the current scene (that is, are more relevant to the event model) be reactivated more than scenes with a different context? To assess this, the full text of each experimental stimulus was analyzed: descriptions of the scenes (including transcript of dialogues) in the Sherlock movie[55], and the text of the 21st year story. Under the assumption that word cooccurrences between scenes would represent context similarity, for each experimental stimulus separately, the words comprising each scene were used to construct a Bag-of-Words model, as implemented in Matlab. The Bag-of-Words model listed all words across each full experimental stimulus (excluding stop words, such as 'the,' 'a' and so on, that have no specific semantic meaning), and counted the number of occurrences of each word in each scene. These scene-specific occurrence vectors were then transformed into probability vectors. Thus, for each scene, a probability vector the length of all words in the experimental narrative was created (see illustration in Fig. 1d, bottom right). To measure context similarities between scenes, the Jensen–Shannon distance was calculated between each two scene-vectors, and the resulting between-scene distance matrix was converted to a similarity matrix. This matrix was meant to be correlated with the neural event boundary × scene similarity matrix.

However, visual inspection of the Bag-of-Words scene similarity matrices of both datasets revealed a time-related bias of similarity values (Supplementary Fig. 6a,c). Bag-of-Words scene similarities were significantly correlated with time (the serial position of scenes) in the Sherlock data ($r_{(901)} = 0.69$; $P < 0.001$). This bias was related to the first 15 scenes of the movie, which were either very short (and thus had unreliable Bag-of-Words representations) or less related to the overall narrative (containing noncentral characters/events, which resulted in unique Bag-of-Words representations). The first 15 scenes were indeed less similar to all other scenes, compared with the between-scene similarities of the remaining scenes ($t_{(901)} = -25.9$, $P < 0.001$; Supplementary Fig. 6a). Thus, correlating the Bag-of-Words scene-similarity matrix with the neural similarity matrix would capture time-related effects, rather than the more subtle modulation of reactivation by semantic context. A similar time-related bias was apparent across the congruent entries of the 21st year Bag-of-Words scene-similarity matrix ($r_{(763)} = -0.08$; $P = 0.03$). This bias was related to the merging of the two separate narratives in the last 15 scenes of the story. By definition, the congruent entries of each separate narrative would be more similar to each other (as they contain very similar words) compared with the similarities between the entries of the joint narrative and each of the separate narratives (since the joint narrative contains a mixture of words from both narratives, and would therefore be less similar to each separate narrative). The between-scene similarities of the last 15 scenes (which are, by definition, congruent with all other scenes) were indeed lower compared with the congruent entries of the between-scene similarities of the first 30 scenes ($t_{(553)} = -10.22$; $P < 0.001$; Supplementary Fig. 6c). To prevent these time-related biases from masking true experimental effects, the first 15 scenes of the Sherlock dataset and the last 15 scenes of the 21st year dataset were excluded from the context similarity matrices. This procedure successfully eliminated the time-biases ($r_{(433)} = 0.05/0.02$; $P = 0.3/0.7$, for Sherlock/21st year; Supplementary Fig. 6b,d). For transparency, a whole-brain Bag-of-Words analysis was also conducted using the full, time-biased Bag-of-Words context similarity matrices.

Thus, for the Sherlock dataset, the following analysis was performed: for each participant and sphere, a neural event boundary × scene similarity matrix was computed using either a within- or between-participants analysis, as previously described. A context scene-similarity matrix was also calculated. Due to the variable scene length in the Sherlock movie (Table 1), which could affect the context similarity values of short scenes (with very sparse vectors), a third matrix indicating the minimal number of words in each scene-pair was also created. Next, the past part of the neural similarity matrix was correlated with the past part of the context similarity matrix, and the future part of the neural similarity matrix was correlated with the future part of the context similarity matrix (which is symmetric to the past part of the same matrix). In both computations, the contribution of the minimal-word matrix was partialled out of the correlation. Finally, the difference between the Fisher-transformed past and future correlation coefficients was computed, cancelling out any effect of scene similarities, which is symmetric in time. This difference value was assigned to the center of the sphere. Single-subject and group maps were created as described for the reactivation analysis. However, given the clear directionality of the experimental hypothesis (similarity in context will induce more reactivation), one-tailed *P* values were computed.

The same analysis could not be performed for the 21st year dataset, due to its experimental design. Due to the alternating narratives across the first 30 scenes, a checkerboard pattern of similarity was captured by the Bag-of-Words model (Supplementary Fig. 6c,d). Similarly to the previously described time bias, correlating the context scene-similarity matrix with the neural similarity matrix would capture the robust congruency effect rather than the more subtle modulation of reactivation by sematic context in the congruent scenes. For this reason, rather than using the main effect of time as the basis for the neural reactivation index, a time × congruency interaction was used. Specifically, for each participant and sphere, event boundary × scene similarity matrices were calculated as described above. Next, separately for the past and future parts of each neural similarity matrix, the congruent entries were correlated with the corresponding entries in the context similarity matrix, and the incongruent entries were correlated with corresponding entries in the context similarity matrix. Finally, the difference between the Fisher-transformed congruent and incongruent correlations of the past part of the matrix were contrasted with the difference between the Fisher-transformed congruent and incongruent correlations of the future part of the matrix. This index was assigned to the center of each sphere. Single-subject and one-tailed group maps were created as described for the reactivation analysis.

ROI analysis. To test whether the same regions that showed a reactivation effect also showed modulation of this reactivation by semantic context, an ROI analysis was performed. This analysis was confined to the precuneus/retrosplenial cortex, since this was the only region that showed a reproducible context-modulation effect across datasets and analysis methods in the whole-brain analysis.

The same precuneus/retrosplenial cortex ROIs that were defined from the reactivation analysis group maps were used. Since these ROIs were based on the reactivation analysis, they were independent with regards to context-modulation analysis. Thus, the ROIs defined from the reactivation analysis for each dataset (Sherlock and 21st year) and analysis method (within and between participants) were used to test the modulation of reactivation by semantic context for the same dataset and analysis method. Specifically, for each participant, the relevant (dataset- and analysis-specific) ROI was used to extract event boundaries and scene representations. Neural similarity matrices were constructed, and correlated with a context similarity matrix, as previously described. This resulted in four sets of values derived from single participants (Sherlock and 21st year participants in the within- and between-participant analysis).

Group statistics were carried out using permutation tests. Each set of single-subject values was submitted to a one-sample *t*-test, and the group level *t*-value was defined as the statistic for the permutation test. Under the null hypothesis of no contextual modulation of reactivation, the sign of single-subject values could be flipped. Thus, for each dataset/analysis method, values of single participants were flipped randomly, and a random group *t*-value was calculated. This procedure was repeated 1,000 times, resulting in 1,000 random *t* values that constructed the null distribution. The position of the true test statistic in relation to the null distribution was used to determine the one-tailed *P* value of the test.

To test the reproducibility of the contextual modulation of reactivation in the precuneus/retrosplenial cortex across datasets, the two dataset-specific *P* values resulting from the above described permutation tests were meta-analysed using Fisher's method. This was done separately for each of the analysis methods (within and between participants).

Mega analyses

A pooled whole-brain analysis across datasets was conducted using FSL's randomize. Single-subject maps of all participants were mega-analyzed using a second-level generalized linear model design, where each dataset was modelled with a dummy regressor. Resulting mega-analysis maps were thresholded and corrected for multiple comparisons, as described for the within-dataset group analyses.

A mega-analysis of the context specificity of reactivation in the precuneus/retrosplenial cortex was also conducted using a permutation test. The only difference to the permutation test previously described for the within-dataset analysis was the inclusion of dummy regressors for each dataset.

Autocorrelations in within-participant analyses

Calculating correlations between representations of single TRs (event boundaries) and other timepoints of the same fMRI signal is problematic, since these representations are embedded in 1/f noise (slow trends in the time domain, which contribute to the lower frequencies of the power spectrum)[69]. This noise is typically removed by applying a linear high-pass filter, which removes from each TR the commonalities with its weighted temporal neighborhood. By doing so, the correlations between each TR and its environment are removed, thus 'flattening' the linear trend induced by the 1/f noise (Supplementary Fig. 8a). However, this procedure changes the autocorrelation structure of fMRI data: before filtering, each TR has positive correlations with neighboring TRs that slowly decline in time, but high-pass filtering induces a more complex structure of alternating positive and negative correlations that decline in time (Supplementary Fig. 8b). Thus, correlations between event boundaries and scenes will have a temporal structure, which is symmetric in time, because event boundaries are symmetric with regards to past and future scenes. However, since the scenes that immediately precede event boundaries are removed from the similarity matrices, past scenes are further away from event boundaries compared with future scenes. This entails that the temporal structure introduced by autocorrelations will be asymmetric with respect to past and future scenes, thus biasing the reactivation index.

Moreover, this temporal bias will also interact with any temporal asymmetries introduced by the structure of the experimental stimulus. Datasets based on experimental stimuli with short scenes (for example, the Sherlock dataset) will be impacted more heavily by autocorrelation biases compared with datasets based on experimental stimuli with longer scenes (for example, the 21st year dataset). The relative position of short scenes will also determine the magnitude of bias (series of adjacent short scenes will be heavily impacted). For example, the beginning of the movie Sherlock has a cluster of short scenes and, therefore, on average, event boundaries will be closer in time to past scenes than to future scenes. In addition, inter-scene intervals, such as those in the 21st year story, entail that event boundaries are closer to their past (containing) scenes, compared with future (noncontinuous) scenes. Finally, datasets based on experimental stimuli with a time-alternating structure (for example, the 21st year dataset) will have a different bias compared with datasets devoid of such a structure (for example, the Sherlock dataset), as discussed below. Therefore, to determine what would be the most suitable unbiased statistic for each dataset (Sherlock and 21st year), simulations based on the unique characteristics of each dataset and its related experimental stimulus were employed.

Note again that these autocorrelation biases do not exist in the between-participant analysis, where the representations of event boundaries and scenes are extracted from different brains (independent signals). The fact that the between-participant effects cleanly replicate the within-participant effects should increase confidence that autocorrelations have been suitably accounted for using the approach described in the following sections.

Reactivation index. To control for the autocorrelation bias, adjacent scenes on both sides of event boundaries, which are heavily impacted by the bias, must be removed from our analyses. To assess the number of scenes that must be removed, ten 'single-subject' simulated datasets of 120 voxels with a 1/f power spectrum were created and high-pass filtered. These datasets were created based on either the characteristics of the Sherlock experiment (TR = 1.5, scan length = 1,976 TRs, high-pass filter of 140 s) or the 21st year experiment (TR = 1.5, scan length = 2,242 TRs, high-pass filter of 120 s). In each simulated dataset, representations of event boundaries and scenes were extracted based on the specific timings of scenes in the experimental stimuli. Representations of event boundaries and scenes were then correlated within each simulated dataset, yielding ten random similarity matrices. These ten matrices were used to calculate ten random reactivation indices, which were then averaged into a group index. The ten similarity matrices were later averaged across the simulated datasets to create a group similarity matrix. This procedure was repeated 10,000 times, resulting in 10,000 group reactivation indices and 10,000 group similarity matrices. The group similarity matrices were further averaged for display purposes.

As evident in the upper left panels of Supplementary Fig. 8c,d, both the Sherlock and 21st year averaged similarity matrices showed artefactual correlations around their main diagonals (the missing rows and columns in the Sherlock matrix correspond to excluded scenes and event boundaries, as described above). The lower left panels of Supplementary Fig. 8c,d demonstrate that these artefactual correlations were reflected in positively biased distributions of the reactivation indices. However, removing ten diagonals from the random Sherlock matrices and six diagonals from the random 21st year matrices removed the artefactual correlations (upper right panels of Supplementary Fig. 8c,d) and centred the reactivation index distributions on 0 (lower right panels of Supplementary Fig. 8c,d), thus removing autocorrelation-related biases from our analysis.

Note that removing diagonals from the similarity matrix means discarding of scenes that precede or follow each event boundary (5

and 3 scenes on each side of event boundaries in the Sherlock and 21st year datasets, respectively). An equal number of diagonals from the past and future parts of the similarity matrix must be discarded, since high-pass filtering has a symmetric effect on timepoints that precede of follow each event boundary. For this same reason, the main diagonal must belong to the past part of the similarity matrix, since it contains the scenes immediately preceding each event boundary. Removal of n diagonals from the similarity matrix therefore entails the removal of $n/2$ diagonals from the upper triangular part of the matrix (future part), and the removal of the main diagonal and additional $n/2 - 1$ diagonals from the lower part of the matrix (past part). The reactivation index would therefore take the following form:

$$\frac{(n-d)(n-d+1)}{2}\sum_{i=d+1}^{n}\sum_{j=1}^{i-d}M_{ij} - \frac{(n-d)(n-d-1)}{2}\sum_{i=1}^{n-d-1}\sum_{j=i+d+1}^{n}M_{ij}$$

Where M is an $n \times n$ matrix, i and j are row and column indices, respectively, and d is the number of diagonals removed from the upper triangular part of the matrix.

Though our analyses create a slightly unbalanced partition of the similarity matrix (more past than future entries), the simulations presented in the lower right panels of Supplementary Fig. 8c,d demonstrate that this imbalance does not bias the reactivation index distribution.

To assess whether stable scene patterns may also introduce some bias into the analysis, the simulations described above were repeated while adding in random scene representations drawn from a uniform distribution. These added representations were given weights of 0.1, 0.5, 1 and 1.5 with regards to the averaged scene representations derived from the random 1/f signals. As demonstrated in Supplementary Fig. 9 (for weights 0.5 and 1), the higher the weight, the more the added patterns masked the autocorrelation bias, as scene representations no longer reflected solely the 1/f signals from which they arose. Yet the removal of diagonals was still needed to center the reactivation index distribution on zero. In other words, these uncorrelated scene patterns simply added unexplained variance to the analysis, suggesting that it is clearly not mere scene representations that cause the observed bias.

To investigate whether autocorrelations between scene representations could cause the bias, simulations were repeated while adding in weighted scene patterns drawn from a 1/f distribution. As demonstrated in Supplementary Fig. 10, adding in autocorrelated scene representations did not bias the analysis, but only masked the time-related autocorrelation bias, as scene patterns no longer solely reflect the originating 1/f signals (and again, removal of diagonals was needed to control for the bias). This is because scene autocorrelations are a function of scene order and not of time. This means they are perfectly symmetric with regards to event boundaries and will get subtracted out by our measure.

To assess whether a congruency structure, as in the 21st year dataset, could underlie the bias, random template scene representations were drawn for narrative A and for narrative B of the story. These representations were added to the simulated scene representations in an interleaved manner, and were added Gaussian noise to create correlations between congruent scene representations. As demonstrated in Supplementary Fig. 11, adding in these weighted scene patterns again masked the observed bias, overruling the possibility that mere congruency structure caused the autocorrelation bias.

However, the congruency structure of the 21st year dataset does interact with the time-related autocorrelation bias, which prevented the use of a time (past/future) × congruency (same/different narrative) interaction in the calculation of the reactivation index. As previously described, and schematically illustrated in Supplementary Fig. 12a, the first 30 scenes of the 21st year datasets consisted of alternating narratives, that later converge. The similarity matrices calculated for this dataset would therefore contain alternating entries of congruent (same narrative) or incongruent (different narrative) correlations between event boundaries and scenes. These entries are arranged by diagonals, which create an asymmetric congruency structure with regards to event boundaries. For example, the main diagonal contains correlations between event boundaries and their immediately preceding scenes, which are congruent. The superdiagonal contains correlations between event boundaries and their immediately following scenes, which are incongruent. However, autocorrelations have a symmetric effect in time. In raw data, event boundaries are more correlated to closer points in time, and are hence more correlated to congruent scenes in the past and to incongruent scenes in the future (this trend flips after applying a high-pass filter). As a result, any congruency-based analysis, such as a time × congruency interaction, will suffer from autocorrelation biases. This bias cannot be fixed by removing diagonals from the analysis, since an equal number of diagonals must be removed from the past/future parts of the matrix, which will maintain an asymmetric congruency structure in time. Indeed, when repeating the simulations of our analysis with random 1/f data, as described above, but this time defining the reactivation index as a time × congruency interaction, the removal of the diagonals failed to remove the autocorrelation artefact from the reactivation index distribution (the distribution is positively biased, Supplementary Fig. 12b). For this reason, a reactivation index that captures the main effect of time (past–future) was chosen for both the Sherlock and 21st year datasets. For consistency, the same reactivation index was also used in the main between-participant analyses.

Control timepoints. As explained in Time-specificity of reactivation, the choice of an appropriate timepoint to serve as a control is non-trivial, due to the influence of autocorrelations. Event boundaries are, by definition, positioned between two adjacent scenes. Due to this position, the effect of high-pass filtering on the event boundary is symmetric with respect to time, since the local temporal environment of the event boundary contains equal timepoints from past and future scenes (this is true for the Sherlock dataset, where no time-gaps exist between scenes. This is marginally true for the 21st year dataset, where inter-scene intervals were inserted, making event boundaries slightly closer to past scenes than to future scenes). However, a control timepoint that precedes an event boundary will be much closer to the past period (as defined by the boundary) and a control timepoint that follows an event boundary will be much closer to the future period (Supplementary Fig. 13a). In both cases, high-pass filtering will induce unbalanced temporal autocorrelations between the control timepoint and past/future scenes, and will therefore introduce a temporal bias into our analyses.

To illustrate this point, simulations of random high-pass filtered 1/f data were repeated for the two experiments, as described above, but this time, past (ten timepoints before event boundaries) and future (ten timepoints after event boundaries) control timepoints were added to the analysis. Representations were extracted from each of these timepoints separately (event boundaries, past/future controls). These representations were then correlated with scene representations to construct similarity matrices, diagonals were discarded as described above and reactivation indices were computed. As depicted in Supplementary Fig. 13b, and as previously demonstrated, the distribution of the reactivation index at event boundaries was centred on zero in the datasets of both experiments. However, the distributions of reactivation indices of both the past and future control timepoints were not centered on zero, indicating an autocorrelation artefact that interacts with time (Supplementary Fig. 13c,d). The results of these simulations indicate that contrasting the reactivation indices of the event boundaries with those of either past or future control timepoints would introduce biases into our analyses. However, since high-pass filtering

is symmetric in time, using both control timepoints, which are equally distant from the event boundary, would cancel out the temporal bias and create a balanced control. Simulation analyses indeed revealed that contrasting the reactivation index of event boundaries with that of the balanced control yielded distributions centered on zero (Supplementary Fig. 13e), indicating an unbiased analysis.

Context-modulation of reactivation in the 21st year dataset. As demonstrated previously, using a time × congruency interaction-based reactivation index for the 21st year dataset inherently induced biases into the analysis. To evaluate whether the context-modulation analysis would be similarly affected by this bias, simulations were performed. In these simulations, the interaction of correlations between the random neural similarity matrices and the context similarity matrices was computed. This procedure was repeated 1,000 times, constructing a random distribution of interaction values, which showed a negative bias (Supplementary Fig. 12c). Since this bias was in the opposite direction to the experimental hypothesis, it would attenuate rather than enhance any experimental effect. Thus, weaker results were expected to be found for this analysis in the 21st year dataset compared with the Sherlock dataset.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data underlying the findings of this study are available at https://dataspace.princeton.edu/handle/88435/dsp01nz8062179 and https://openneuro.org/datasets/ds002345/versions/1.1.4.

Code availability

Matlab code is available at https://osf.io/crmdb/.

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Author contributions

A.H. conducted all analyses. H.D. assisted with natural language processing analyses. A.H., H.D. and T.E.J.B. interpreted the results. A.H. and T.E.J.B. wrote the manuscript. T.E.J.B. supervised the project.

Competing interests

The authors declare no competing interests.

Additional information

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Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our Editorial Policies and the Editorial Policy Checklist.

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (*n*) for each experimental group/condition, given as a discrete number and unit of measurement
☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided *Only common tests should be described solely by name; describe more complex techniques in the Methods section.*
☒	☐ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's *d*, Pearson's *r*), indicating how they were calculated

Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	no software was used for data collection, as data were obtained from online databases
Data analysis	FSL 6.0.5 (including all embedded tools), Matlab R2018a, code is available at https://osf.io/crmdb/

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio guidelines for submitting code & software for further information.

Data

Policy information about availability of data

All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our policy

This study reports findings from two naturalistic fMRI datasets (see Table 1): The Sherlock dataset[40] downloaded from Princeton University's DataSpace repository[37]; and the 21st year dataset[39], downloaded from the Narratives repository[38].

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	Participants Sherlock. 22 participants (10 female, age range 18-26) were recruited for the original study, of which 17 met the original study's inclusion criteria, and were therefore also analysed in the current study. 21st year. 25 participants (14 female, age range 18-40) were analysed in the original and current study. Gender data has not been collected, no sex/gender-based analyses were performed.
Population characteristics	Participants Sherlock. 22 participants (10 female, age range 18-26) were recruited for the original study, of which 17 met the original study's inclusion criteria, and were therefore also analysed in the current study. 21st year. 25 participants (14 female, age range 18-40) were analysed in the original and current study.
Recruitment	N.A. as data were obtained from online databases
Ethics oversight	N.A. as data were obtained from online databases. The original experimental protocols were approved by the Institutional Review Board of the Princeton University.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	we made used of two freely-available datasets of participants (n1=22, n2=25) who were engaged in the comprehension of a long naturalistic narrative. These two dataset were chosen as they were the only large datasets with long-enough scanning periods of movie data at the time of data analysis. The use of these two datasets allowed the full replication of results.
Data exclusions	Participants Sherlock. 22 participants (10 female, age range 18-26) were recruited for the original study, of which 17 met the original study's inclusion criteria, and were therefore also analysed in the current study. 21st year. 25 participants (14 female, age range 18-40) were analysed in the original and current study.
Replication	we replicated all results across two independent datasets
Randomization	N.A. as this is a one-sample design (only a single group of participants were recruited in each dataset, as there is no experimental manipulation). Data from each dataset was acquired and analyzed separately. When using a pooled analysis, the attribution of datasets was regressed out of the data using dummy regressors.
Blinding	N.A. as this is a one-sample design, there is only one group of participants in each dataset

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study type including whether data are quantitative, qualitative, or mixed-methods (e.g. qualitative cross-sectional, quantitative experimental, mixed-methods case study).*
Research sample	*State the research sample (e.g. Harvard university undergraduates, villagers in rural India) and provide relevant demographic information (e.g. age, sex) and indicate whether the sample is representative. Provide a rationale for the study sample chosen. For studies involving existing datasets, please describe the dataset and source.*
Sampling strategy	*Describe the sampling procedure (e.g. random, snowball, stratified, convenience). Describe the statistical methods that were used to*

Sampling strategy	*predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient. For qualitative data, please indicate whether data saturation was considered, and what criteria were used to decide that no further sampling was needed.*
Data collection	*Provide details about the data collection procedure, including the instruments or devices used to record the data (e.g. pen and paper, computer, eye tracker, video or audio equipment) whether anyone was present besides the participant(s) and the researcher, and whether the researcher was blind to experimental condition and/or the study hypothesis during data collection.*
Timing	*Indicate the start and stop dates of data collection. If there is a gap between collection periods, state the dates for each sample cohort.*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, provide the exact number of exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Non-participation	*State how many participants dropped out/declined participation and the reason(s) given OR provide response rate OR state that no participants dropped out/declined participation.*
Randomization	*If participants were not allocated into experimental groups, state so OR describe how participants were allocated to groups, and if allocation was not random, describe how covariates were controlled.*

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	*Briefly describe the study. For quantitative data include treatment factors and interactions, design structure (e.g. factorial, nested, hierarchical), nature and number of experimental units and replicates.*
Research sample	*Describe the research sample (e.g. a group of tagged Passer domesticus, all Stenocereus thurberi within Organ Pipe Cactus National Monument), and provide a rationale for the sample choice. When relevant, describe the organism taxa, source, sex, age range and any manipulations. State what population the sample is meant to represent when applicable. For studies involving existing datasets, describe the data and its source.*
Sampling strategy	*Note the sampling procedure. Describe the statistical methods that were used to predetermine sample size OR if no sample-size calculation was performed, describe how sample sizes were chosen and provide a rationale for why these sample sizes are sufficient.*
Data collection	*Describe the data collection procedure, including who recorded the data and how.*
Timing and spatial scale	*Indicate the start and stop dates of data collection, noting the frequency and periodicity of sampling and providing a rationale for these choices. If there is a gap between collection periods, state the dates for each sample cohort. Specify the spatial scale from which the data are taken*
Data exclusions	*If no data were excluded from the analyses, state so OR if data were excluded, describe the exclusions and the rationale behind them, indicating whether exclusion criteria were pre-established.*
Reproducibility	*Describe the measures taken to verify the reproducibility of experimental findings. For each experiment, note whether any attempts to repeat the experiment failed OR state that all attempts to repeat the experiment were successful.*
Randomization	*Describe how samples/organisms/participants were allocated into groups. If allocation was not random, describe how covariates were controlled. If this is not relevant to your study, explain why.*
Blinding	*Describe the extent of blinding used during data acquisition and analysis. If blinding was not possible, describe why OR explain why blinding was not relevant to your study.*

Did the study involve field work? ☐ Yes ☐ No

Field work, collection and transport

Field conditions	*Describe the study conditions for field work, providing relevant parameters (e.g. temperature, rainfall).*
Location	*State the location of the sampling or experiment, providing relevant parameters (e.g. latitude and longitude, elevation, water depth).*
Access & import/export	*Describe the efforts you have made to access habitats and to collect and import/export your samples in a responsible manner and in compliance with local, national and international laws, noting any permits that were obtained (give the name of the issuing authority, the date of issue, and any identifying information).*
Disturbance	*Describe any disturbance caused by the study and how it was minimized.*

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☐	☒ MRI-based neuroimaging

Antibodies

Antibodies used — *Describe all antibodies used in the study; as applicable, provide supplier name, catalog number, clone name, and lot number.*

Validation — *Describe the validation of each primary antibody for the species and application, noting any validation statements on the manufacturer's website, relevant citations, antibody profiles in online databases, or data provided in the manuscript.*

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s) — *State the source of each cell line used and the sex of all primary cell lines and cells derived from human participants or vertebrate models.*

Authentication — *Describe the authentication procedures for each cell line used OR declare that none of the cell lines used were authenticated.*

Mycoplasma contamination — *Confirm that all cell lines tested negative for mycoplasma contamination OR describe the results of the testing for mycoplasma contamination OR declare that the cell lines were not tested for mycoplasma contamination.*

Commonly misidentified lines (See ICLAC register) — *Name any commonly misidentified cell lines used in the study and provide a rationale for their use.*

Palaeontology and Archaeology

Specimen provenance — *Provide provenance information for specimens and describe permits that were obtained for the work (including the name of the issuing authority, the date of issue, and any identifying information). Permits should encompass collection and, where applicable, export.*

Specimen deposition — *Indicate where the specimens have been deposited to permit free access by other researchers.*

Dating methods — *If new dates are provided, describe how they were obtained (e.g. collection, storage, sample pretreatment and measurement), where they were obtained (i.e. lab name), the calibration program and the protocol for quality assurance OR state that no new dates are provided.*

☐ Tick this box to confirm that the raw and calibrated dates are available in the paper or in Supplementary Information.

Ethics oversight — *Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Animals and other research organisms

Policy information about studies involving animals; ARRIVE guidelines recommended for reporting animal research, and Sex and Gender in Research

Laboratory animals — *For laboratory animals, report species, strain and age OR state that the study did not involve laboratory animals.*

Wild animals	*Provide details on animals observed in or captured in the field; report species and age where possible. Describe how animals were caught and transported and what happened to captive animals after the study (if killed, explain why and describe method; if released, say where and when) OR state that the study did not involve wild animals.*
Reporting on sex	*Indicate if findings apply to only one sex; describe whether sex was considered in study design, methods used for assigning sex. Provide data disaggregated for sex where this information has been collected in the source data as appropriate; provide overall numbers in this Reporting Summary. Please state if this information has not been collected. Report sex-based analyses where performed, justify reasons for lack of sex-based analysis.*
Field-collected samples	*For laboratory work with field-collected samples, describe all relevant parameters such as housing, maintenance, temperature, photoperiod and end-of-experiment protocol OR state that the study did not involve samples collected from the field.*
Ethics oversight	*Identify the organization(s) that approved or provided guidance on the study protocol, OR state that no ethical approval or guidance was required and explain why not.*

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Clinical data

Policy information about clinical studies

All manuscripts should comply with the ICMJE guidelines for publication of clinical research and a completed CONSORT checklist must be included with all submissions.

Clinical trial registration	*Provide the trial registration number from ClinicalTrials.gov or an equivalent agency.*
Study protocol	*Note where the full trial protocol can be accessed OR if not available, explain why.*
Data collection	*Describe the settings and locales of data collection, noting the time periods of recruitment and data collection.*
Outcomes	*Describe how you pre-defined primary and secondary outcome measures and how you assessed these measures.*

Dual use research of concern

Policy information about dual use research of concern

Hazards

Could the accidental, deliberate or reckless misuse of agents or technologies generated in the work, or the application of information presented in the manuscript, pose a threat to:

No	Yes	
☐	☐	Public health
☐	☐	National security
☐	☐	Crops and/or livestock
☐	☐	Ecosystems
☐	☐	Any other significant area

Experiments of concern

Does the work involve any of these experiments of concern:

No	Yes	
☐	☐	Demonstrate how to render a vaccine ineffective
☐	☐	Confer resistance to therapeutically useful antibiotics or antiviral agents
☐	☐	Enhance the virulence of a pathogen or render a nonpathogen virulent
☐	☐	Increase transmissibility of a pathogen
☐	☐	Alter the host range of a pathogen
☐	☐	Enable evasion of diagnostic/detection modalities
☐	☐	Enable the weaponization of a biological agent or toxin
☐	☐	Any other potentially harmful combination of experiments and agents

ChIP-seq

Data deposition

☐ Confirm that both raw and final processed data have been deposited in a public database such as GEO.

☐ Confirm that you have deposited or provided access to graph files (e.g. BED files) for the called peaks.

Data access links *May remain private before publication.*	*For "Initial submission" or "Revised version" documents, provide reviewer access links. For your "Final submission" document, provide a link to the deposited data.*
Files in database submission	*Provide a list of all files available in the database submission.*
Genome browser session (e.g. UCSC)	*Provide a link to an anonymized genome browser session for "Initial submission" and "Revised version" documents only, to enable peer review. Write "no longer applicable" for "Final submission" documents.*

Methodology

Replicates	*Describe the experimental replicates, specifying number, type and replicate agreement.*
Sequencing depth	*Describe the sequencing depth for each experiment, providing the total number of reads, uniquely mapped reads, length of reads and whether they were paired- or single-end.*
Antibodies	*Describe the antibodies used for the ChIP-seq experiments; as applicable, provide supplier name, catalog number, clone name, and lot number.*
Peak calling parameters	*Specify the command line program and parameters used for read mapping and peak calling, including the ChIP, control and index files used.*
Data quality	*Describe the methods used to ensure data quality in full detail, including how many peaks are at FDR 5% and above 5-fold enrichment.*
Software	*Describe the software used to collect and analyze the ChIP-seq data. For custom code that has been deposited into a community repository, provide accession details.*

Flow Cytometry

Plots

Confirm that:

☐ The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).

☐ The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).

☐ All plots are contour plots with outliers or pseudocolor plots.

☐ A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	*Describe the sample preparation, detailing the biological source of the cells and any tissue processing steps used.*
Instrument	*Identify the instrument used for data collection, specifying make and model number.*
Software	*Describe the software used to collect and analyze the flow cytometry data. For custom code that has been deposited into a community repository, provide accession details.*
Cell population abundance	*Describe the abundance of the relevant cell populations within post-sort fractions, providing details on the purity of the samples and how it was determined.*
Gating strategy	*Describe the gating strategy used for all relevant experiments, specifying the preliminary FSC/SSC gates of the starting cell population, indicating where boundaries between "positive" and "negative" staining cell populations are defined.*

☐ Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.

Magnetic resonance imaging

Experimental design

Design type	naturalistic tasks (movie watching / listening to a story)

Design specifications	Sherlock. Here we analysed data from participants who watched the first 48 minutes of Episode 1 of the BBC's television series "Sherlock". The plot had a continuous structure, with varying scene-lengths (median 51 seconds, range 12 - 184.5 seconds). Scene timings were taken from the original study, where they were manually defined based on major shifts in the narrative[40]. The experimental stimulus was divided into two segments of 23/25 minutes, and a 30 seconds audiovisual cartoon (unrelated to the main stimulus) was presented prior to each segment. 21st year. Participants listened to a 56 minute-long story, which was segmented into scenes by design. These scenes had relatively constant lengths (median 67.5 seconds, range 60 - 84 seconds), and were separated by 4.5-6 seconds of silence. The narrative was composed of two unrelated storylines that alternated during the first 30 story-scenes, and then merged into a unified storyline during the last 15 story-scenes.
Behavioral performance measures	N.A. passive task

Acquisition

Imaging type(s)	functional
Field strength	3T
Sequence & imaging parameters	Functional images were acquired using a T2*-weighted echo planar imaging (EPI) pulse sequence (TR 1500 ms, TE 28 ms, flip angle 64°), each volume comprising 27 slices of 4 mm thickness, in-plane resolution 3 × 3 mm2, FOV 192×192 mm2), with ascending interleaved acquisition.
Area of acquisition	whole brain

Diffusion MRI ☐ Used ☒ Not used

Preprocessing

Preprocessing software	FSL 6.0.5 (includes all preprocessing algorithms)
Normalization	Functional images were aligned to structural images initially using FMRIB's Linear Image Registration Tool (FLIRT)[57, 59] and then optimized using Boundary-Based Registration (BBR)[60]. Structural images were transformed into MNI space using FMRIB's Nonlinear Image Registration Tool (FNIRT) and the resulting warp fields were applied to the functional images.
Normalization template	MNI152 standard-space T1-weighted average structural template image
Noise and artifact removal	Tissue-type segmentation was carried out using FAST[61] to create white matter/CSF nuisance masks. To avoid the inclusion of grey matter voxels in these nuisance masks, these masks included only voxels identified as white matter/CSF with probability of 1, restricted by an anatomically-based bounding box (-44<x<42, -84<y<42, -4<z<34 for white matter; -42<x<38, -64<y<38, -22<z<28 for CSF), and eroded by a sphere of radius 5 and 2 mm for white-matter and CSF, respectively[62]. White-matter and CSF time-courses were extracted for each functional scan, and their contribution to the BOLD signal, as well as the contribution of motion parameters, was later removed.
Volume censoring	no volumes were censored

Statistical modeling & inference

Model type and settings	RSA, group statistics were calculated based on one-sample t-tests and permutation tests
Effect(s) tested	we tested whether past scenes are reactivated at event boundaries

Specify type of analysis: ☐ Whole brain ☐ ROI-based ☒ Both

Statistic type for inference (See Eklund et al. 2016)	cluster-wise. Group maps were corrected for multiple comparisons using a permutation test, using the null distribution of the maximal cluster mass, as implemented in FSL's randomize, thresholded at p<0.005.
Correction	Group maps were corrected for multiple comparisons using a permutation test, using the null distribution of the maximal cluster size, as implemented in FSL's randomize, thresholded at p<0.005.

Models & analysis

n/a	Involved in the study
☒	☐ Functional and/or effective connectivity
☒	☐ Graph analysis
☒	☐ Multivariate modeling or predictive analysis

Functional and/or effective connectivity	*Report the measures of dependence used and the model details (e.g. Pearson correlation, partial correlation, mutual information).*
Graph analysis	*Report the dependent variable and connectivity measure, specifying weighted graph or binarized graph, subject- or group-level, and the global and/or node summaries used (e.g. clustering coefficient, efficiency, etc.).*
Multivariate modeling and predictive analysis	*Specify independent variables, features extraction and dimension reduction, model, training and evaluation metrics.*